# Anatomy-Guided Masking for I-JEPA Representation Learning on Retinal OCT

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## Abstract

Anatomy-guided masking is motivated by concentrating representation learning on informative tissue, but prediction also requires useful visible context. We study six I-JEPA masking-policy implementations on volumetric retinal OCT, using a 2D B-scan encoder and a frozen volume-level linear probe for glaucoma classification (FairVision, $N$=3000). The encoders descend from a shared checkpoint, with one continuation per policy. Segmenter-guided rectangular target placement improves AUC over uniform placement by $+0.0120$ at epoch 50. A segmentation-free intensity-centroid policy improves it by $+0.0109$ at epoch 100. Both policies have positive paired differences from uniform placement at all three evaluated epochs. Their difference is unresolved at epochs 50 and 75; the centroid policy is higher at epoch 100, which does not establish equivalence between the methods. Greater tissue purity or coverage is not consistently beneficial across the other implementations: an early, incompletely documented anatomy-shaped continuation has higher AUC than its rectangle comparator, whereas later shaped and coverage-constrained policies do not consistently improve the uniform baseline. Auditing delivered masks reveals changes in visible context, target area and loss weighting, including target truncation after coverage-based placement. These comparisons therefore do not isolate the effect of anatomical coverage. Background-feature diagnostics further motivate distinguishing target content from information retained in context, without establishing a causal mechanism. The results support tissue-directed placement in these runs, not a ranking over retrainings: the test split was repeatedly inspected, and subgroup and low-label analyses remain exploratory.

## 1 Introduction

Self-supervised learning has become the default route to representation quality in medical imaging, where expert annotation is expensive and pathology is rare [Azizi et al., 2021, Zhou et al., 2023, Qiu et al., 2024]. Joint-embedding predictive architectures such as I-JEPA [Assran et al., 2023] learn by predicting the *representation* of masked target blocks from a visible context block, avoiding pixel reconstruction and its bias toward high-frequency detail [He et al., 2022, Xie et al., 2022].

I-JEPA samples its targets geometrically: rectangles placed uniformly at random. In natural images this is defensible, since informative content is spread broadly across the frame. In a retinal OCT B-scan it is not obviously so. A large fraction of the image is dark vitreous and background. The diagnostic signal for glaucoma is concentrated in a thin band of retinal tissue. This motivates an intuitive hypothesis, one that several recent methods adopt [Li et al., 2022, 2024, Wang et al., 2025, Shi et al., 2022]: *direct the predictive objective at clinically meaningful anatomy rather than at arbitrary image regions.*

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Selecting targets and retaining context are coupled decisions. Hiding more tissue can increase the amount to predict while removing the evidence needed to predict it. Random targets can also include background whose latent features depend on surrounding tissue. We therefore examine anatomy-directed target selection, encoder-visible context and background supervision jointly, rather than assume that increasing anatomical coverage must help.

We compare six masking-policy implementations, from uniformly placed rectangles to tissue-directed rectangles and anatomy-shaped targets. The encoders descend from a shared pretrained checkpoint, but implementation, collation and historical continuation details differ. We use matched frozen-probe comparisons where available and measure the masks delivered to the model, rather than infer them from configuration names.

The tissue-directed rectangle policies outperform uniform placement in the observed continuations. The centroid policy uses image intensity rather than a segmentation model and has the highest epoch-100 point estimate. The anatomy-shaped results are mixed, including a positive early comparison. Neither these outcomes nor the coverage-constrained arm isolates how much anatomy should be hidden: target area, retained context and the number of supervised target tokens also change. This motivates an implementation audit before a stronger claim about the masking mechanism.

**Contributions.**

- A six-policy study of target selection for OCT representation learning, with a shared encoder ancestry and matched frozen-probe comparisons (every probe fitted is listed in Appendix A).
- Positive run-level results for segmenter-guided and segmentation-free rectangle placement, with mixed outcomes for more anatomically specific implementations.
- An audit of delivered target and context geometry that identifies why the tested implementations do not isolate anatomical coverage. Additional background, low-label and subgroup analyses delimit the interpretation.

## 2 Related work

**Masked image modelling and JEPAs.** Masked autoencoders reconstruct pixels [He et al., 2022, Xie et al., 2022]; BEiT [Bao et al., 2022] and iBOT [Zhou et al., 2022] predict discrete tokens; data2vec [Baevski et al., 2022] and I-JEPA [Assran et al., 2023] predict latent representations. Video and 3D extensions follow the same template [Bardes et al., 2024, Chen et al., 2023b, Taleb et al., 2020]. Other JEPA formulations also change the training objective or prediction structure rather than only the masking distribution [Balestriero and LeCun, 2025, He et al., 2025].

**Region selection and prediction structure.** DSeq-JEPA [He et al., 2025] combines attention-derived region selection with sequential prediction from more salient regions to less salient regions. Its most salient region supplies context rather than being the first prediction target. In its reported factorial ablation, discriminative selection with a flat predictor does not improve on uniform I-JEPA in the reported single-run point estimates, without a significance inference, whereas their joint sequential design does. The regions, saliency source, conditioning and datasets differ from ours; this is not a direct test of retinal anatomy guidance. It demonstrates why successful guided prediction cannot be reduced to the rule “hide more important tissue.” I-JEPA’s own context-scale ablation likewise motivates retaining informative context alongside sufficiently large prediction targets [Assran et al., 2023].

**Informed masking.** A substantial line of work argues that *what* is masked matters. ADIOS [Shi et al., 2022] learns an adversarial masking model; SemMAE [Li et al., 2022] uses learned semantic parts; AttMask [Kakogeorgiou et al., 2022] masks by attention; HardPatches [Wang et al., 2023] and AutoMAE [Chen et al., 2023a] target difficulty. In the medical domain, AnatoMask [Li et al., 2024], AnatomyMAE [Ceballos Arroyo et al., 2026], MSMAE [Mao et al., 2023], and VasoMIM [Huang et al., 2026] explicitly steer masking toward anatomy, and Wang et al. [2025] argue directly for masking clinically relevant regions. Our study examines that premise through run-level comparisons and a delivered-task audit on one task.

**Evidence that informed masking is not uniformly better.** The effect depends on the objective and masking construction. Original MAE uses random patches and reports better performance than its block and grid alternatives [He et al., 2022]. SemMAE's scheduled part-based masking improves its reported linear-probe result, while masking whole parts throughout training does not; its part-map ablation also shows sensitivity to the guide source [Li et al., 2022]. Conversely, anatomy-biased masking helps in a fine-tuned CT aneurysm-detection pipeline that also studies crop selection and distance-map reconstruction [Ceballos Arroyo et al., 2026]. These are different tasks and interventions, not evidence for a universal masking rule. Our OCT comparison asks which conclusions survive a direct audit of target selection and retained context under latent prediction. Additional contextual comparisons are recorded in Appendix N.

**Retinal foundation models, OCT, and fairness.** RETFound [Zhou et al., 2023], VisionFM [Qiu et al., 2024], URFound [Yu et al., 2024], and OCTCube [Liu et al., 2026] pretrain on large retinal corpora. OCT-specific masked modelling is explored by Pissas et al. [2024]. MIRAGE [Morano et al., 2025] provides multimodal OCT segmentation and supplies the anatomical guidance used by several of our arms, and deep glaucoma detection from OCT is established [Maetschke et al., 2019, Ran et al., 2019, Noury et al., 2022]. Performance disparities across demographic groups are widely documented [Seyyed-Kalantari et al., 2021, Larrazabal et al., 2020, Gichoya et al., 2022]. FairVision [Luo et al., 2024a] and Harvard-GF [Luo et al., 2024b] provide OCT data with demographic attributes, and FairMedFM [Jin et al., 2024] and Shi et al. [2025] benchmark fairness for medical foundation models.

## 3 Method: anatomy-guided target selection

We compare five policy families, represented by six historical implementations, with a shared encoder ancestry and a common frozen-probe protocol. The intended intervention is target selection; differences in delivered geometry and historical continuation are reported separately.

### 3.1 Background

I-JEPA samples from the token grid of an image a context block and $M$ target blocks, which need not cover it. A context encoder embeds the visible tokens. A predictor maps that embedding plus positional queries to predicted representations of the target tokens. An exponential-moving-average target encoder supplies the regression targets. With a $16\times16$ token grid the design choice we vary is which of the 256 cells become targets. Figure 1 shows the pipeline. Target blocks are predicted in parallel from the supplied context, with no sequential or causal conditioning on earlier predicted regions.

We distinguish three sets: the patches outside the target union, the candidate context block after target exclusion, and the context tokens retained after collation. Only the last set enters the online encoder. In the geometry tables, mask ratio denotes target-union area divided by grid area, not one minus the fraction of tokens finally visible to the encoder.

### 3.2 The masking-policy implementations

All policies request $M{=}4$ target blocks and one context block. Rectangle policies share nominal scale and aspect-ratio ranges, but placement and collation can change unique target area and encoder-visible context.

**RANDOM (null).** Stock I-JEPA multiblock. Four rectangles placed uniformly at random anywhere in the grid. No image content is consulted.

**ENVELOPE (random-within-retina).** The same nominal rectangle count, scale and aspect-ratio distributions as RANDOM, but rejection-sampled onto the MIRAGE-derived repaired retinal envelope [Morano et al., 2025]. The predicted inner-retina and choroid union is repaired across the unlabeled middle-retina gap in native label space, then transformed jointly with the image. This changes the placement policy, not target shape. Overlap and delivered context are consequences of that intervention, not independently matched controls.

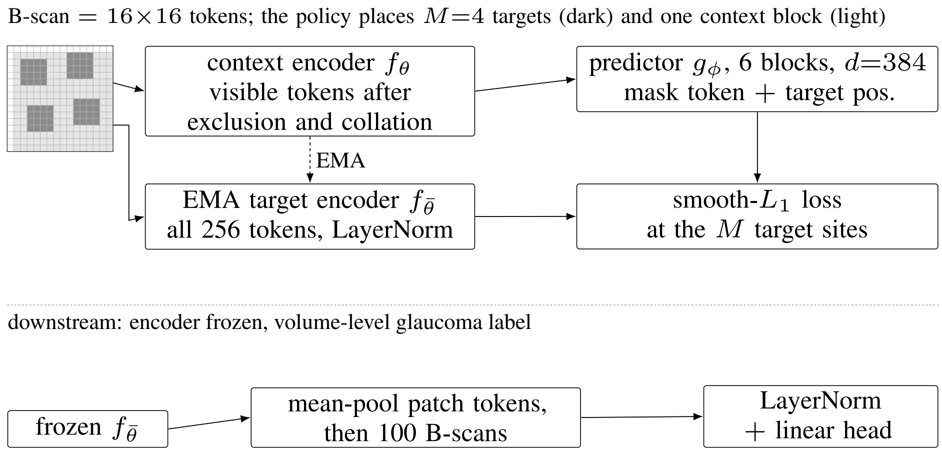


Figure 1: I-JEPA pretraining and frozen-encoder evaluation. The grid is schematic: dark cells are target positions and light cells the candidate context. After exclusion and collation, retained context tokens enter the online encoder. The predictor uses target positions to regress features from the full-image EMA target encoder; target-encoder gradients are stopped. Below the dashed line, the frozen target encoder supplies patch features pooled into one volume-level classification input. Source-verified token-map examples are shown in Figure 4.

**CENTROID (segmentation-free).** A band whose vertical position is located per slice by a per-column intensity-weighted row centroid, smoothed across columns, with no segmentation model and no ground-truth annotation of any kind.

**ANATOMY-V1 / ANATOMY-V2.** Ragged connected components grown on the MIRAGE score map, so target *shape* follows tissue rather than being rectangular. v2 also bridges diagonal adjacencies.

**COVER $f$.** Targets chosen greedily to cover anatomy while leaving a fraction $f$ outside the target union. The historical run uses $f$=0.21. This condition does not guarantee the same fraction survives final encoder-context collation, and post-placement target truncation changes the intended coverage (Appendix G).

Figure 4 shows delivered token sets for the five policy families on one shared view, using ANATOMY-V2 as the shaped-family representative.

### 3.3 Hypotheses

We distinguish three hypotheses, the third of which requires stronger controls than the present policy comparisons provide:

**H1** *Tissue-directed placement helps.* ENVELOPE > RANDOM.
**H2** *Anatomical precision helps further.* anatomy arms > ENVELOPE.
**H3** *Any benefit is due to anatomical targeting, not to incidental changes in mask area, target count, or context size.*

## 4 Experimental setup

**Data.** FairVision [Luo et al., 2024a] glaucoma OCT volumes. Each volume is a 3D scan resampled to 100 B-scans of $256\times256$ after transform. The label is a binary glaucoma diagnosis. Splits are fixed and patient-disjoint as released. Downstream evaluation uses a held-out test split of $N$=3000 volumes (1466 positive / 1534 negative), seed 42, fixed loader order. No test volume is used to fit or select the probe head, which is selected on a separate validation split. The study's choices of policy, checkpoint, analysis and stopping horizon were made after repeated inspection of that same test split (Section 6). Test files are read in deterministic sorted order. The saved label vectors agree across arms, but those vectors alone do not prove case identity because the historical prediction files omit subject identifiers. Paired analyses rely on the documented shared split and loader ordering.

Demographic attributes are used only for the subgroup analysis in Section 5.5 and never as model inputs.

**Pretraining.** Patch-level I-JEPA, ViT-Base, $256\times256$ crops, patch size 16, predictor depth 6, EMA target encoder. The reference fork is the epoch-25 checkpoint, with matched nominal optimiser, learning-rate and weight-decay schedules and effective batch size (512). The checked-in ANATOMY-V1 configuration instead resumes an ENVELOPE epoch-27 checkpoint; its exact historical launch record is not retained, so a direct epoch-25 fork is not independently established for that early result. Guidance is ramped linearly from epoch 25 to 30 and held at full strength thereafter. Within the rectangle family the *only* difference between arms is the mask sampler (Section 3). The anatomy family also differs in collation, and we quantify that in Section 5.

Arms differ in how far they were carried. RANDOM, CENTROID and ENVELOPE reach epoch 100. ANATOMY-V2 reaches epoch 75 (its epoch-75 and epoch-92 probes under an earlier target-precision splice are excluded and replaced by a clean fp32 continuation, Appendix F). We stopped that arm having seen its epoch-75 deficit, which is a selective stopping horizon and a real weakness: a trajectory can recover, and its epoch-100 value is unmeasured, not known to be worse. ANATOMY-V1 has only epoch 30. COVER $f{=}0.21$ reaches epoch 100, with an extra probe at its epoch-73 peak. Epoch 50 is therefore the only epoch at which five policies are simultaneously available, and it carries every cross-policy comparison that involves the anatomy or coverage arms.

**Evaluation.** The frozen encoder is the EMA target branch, not the online context encoder. Per-slice features are mean-pooled over the volume, followed by LayerNorm and a linear head. The head is selected by validation AUC and reported on the held-out test split. We report percentile bootstrap intervals for primary AUCs and paired deltas (10,000 class-stratified subject resamples; fitted heads fixed and each draw shared across arms) and DeLong tests for correlated ROC curves [DeLong et al., 1988], both of which are conditional on the shared case ordering of the documented evaluation loaders. The historical archives contain matching labels but no independent case-ID manifest, as noted above. Each test volume is a distinct subject, so case-level resampling is already subject-level and no cluster correction is required [Obuchowski, 1997, Rutter, 2000]. We follow Maier-Hein et al. [2024] in reporting the operating-point metrics alongside AUC rather than AUC alone (Appendix L).

**Numerical precision.** The probe harness defaults to mixed precision. The RANDOM, CENTROID and ENVELOPE long-horizon probes were fitted under that default (fp16). The ANATOMY-V1, ANATOMY-V2, COVER and ancestor probes were fitted with autocast explicitly disabled (fp32). Protocol is therefore not uniform across probes, so we partition the comparisons: all headline contrasts in Table 1 are drawn from arms sharing both epoch *and* precision, and any contrast that would cross the precision boundary is marked as such and excluded from headline claims. Section 5.6 measures the precision effect.

# 5 Results

The tissue-directed rectangle policies improve AUC in these runs. The causal effect of anatomical coverage is not identified here. Each policy was pretrained once, so the intervals below are sampling error over test subjects, not seed-to-seed variance (Section 6).

## 5.1 Tissue-directed rectangle placement improves the observed AUC

**The observed comparisons are consistent with H1 in these continuations.** ENVELOPE changes rectangle placement relative to the unguided baseline (Section 3). Delivered context also changes ($24.7\%$ to $30.7\%$, Table 5), so this comparison does not isolate location from its consequences for the prediction task. The AUC difference is $+0.0120$ AUC at epoch 50, and CENTROID, a band located from a first-order intensity statistic with no segmentation model anywhere, is worth $+0.0109$ at epoch 100. Every ENVELOPE and CENTROID interval excludes zero at all three epochs, and all six contrasts survive Benjamini–Hochberg correction over that family of nine ($q \leq 0.0299$; Figure 2b, with the full paired inference on all nine precision-matched contrasts in Table 15, Appendix O).

Aiming is not containment: the rectangles are large, so fewer than half of ENVELOPE's masked cells land on tissue (Table 2). Guidance changes where the predictor is repeatedly sent, not what it is

Table 1: Frozen MeanPool probe, FairVision test ($N$=3000, 1466 positive). The reference ancestor is epoch 25 (AUC 0.8487); continuation provenance is described in Section 4. $\Delta$ is versus RANDOM at the *matched* epoch. The upper block is precision-matched (fp16 throughout) and carries every headline claim; the lower block is fp32, each $\Delta$ taken against an fp32 null re-probed at the same epoch. The measured fp16/fp32 gap is at most $2 \times 10^{-4}$ (Appendix P).

| | | epoch 50 | | epoch 75 | | epoch 100 | |
|---|---|---|---|---|---|---|---|
| policy | guidance signal | AUC | $\Delta$ | AUC | $\Delta$ | AUC | $\Delta$ |
| RANDOM | none (null) | 0.8641 | — | 0.8723 | — | 0.8746 | — |
| ENVELOPE | segmenter (location only) | 0.8761 | **+0.0120** | 0.8803 | **+0.0080** | 0.8807 | **+0.0062** |
| CENTROID | intensity centroid, no segmenter | 0.8740 | **+0.0099** | 0.8836 | **+0.0113** | **0.8855** | **+0.0109** |
| ANATOMY-V2 | segmenter (target shape) | 0.8654 | **+0.0013** | 0.8612 | **−0.0111** | — | — |
| COVER $f$=.21 | segmenter (coverage) | 0.8643 | **+0.0002** | 0.8639 | **−0.0084** | 0.8577 | **−0.0168** |

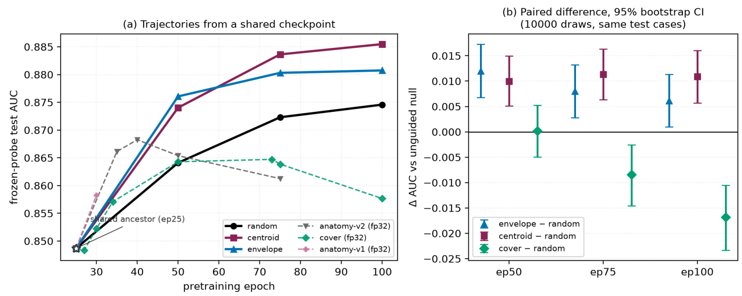


Figure 2: **(a)** Frozen-probe test AUC against pretraining epoch; note the truncated AUC axis. The open star marks the shared epoch-25 ancestor. Solid lines are the precision-matched fp16 family; dashed lines are fp32 arms and must not be compared vertically against the solid lines. **(b)** The actual inference: paired differences against the unguided null on identical test cases, 10,000-resample percentile bootstrap intervals. Every ENVELOPE and CENTROID interval excludes zero; COVER straddles zero at epoch 50 and is negative thereafter. Paired differences account for the covariance of predictions on shared test cases; individual-arm intervals answer a different question.

forbidden to see. Background diagnostics provide a complementary view of the prediction task, not an identified explanation of the AUC differences (Section 5.3).

**The anatomy-shaped results are mixed.** H2 predicts that the anatomy arms exceed ENVELOPE. Tested directly at the matched epoch 50, ANATOMY-V2 sits $-0.0107$ *below* ENVELOPE (CI $[-0.0167, -0.0046]$). At epoch 30, however, ANATOMY-V1 sits $+0.0044$ *above* it (CI $[+0.0009, +0.0078]$), with the early continuation's lineage uncertainty described in Section 4. These are different implementations at different epochs, so we do not read either as settling H2. What the pair rules out is a uniform observed ordering across these implementations, not a causal effect of target shape. Neither ANATOMY-V2 nor COVER is resolved as *worse* than the null at the matched epoch: both sit within a paired interval that comfortably contains zero (Figure 2b). Intervals containing zero do not establish equality or exclude smaller beneficial or adverse effects.

**Carried to epoch 100 the coverage arm degrades.** COVER is the only policy that uses the segmenter to choose *how much* to hide, and carried to the full horizon it produces the study's one *negative* result. It climbs normally to epoch 50, peaks at epoch 73, then *falls*, losing $-0.0071$ from its own peak ($p$<0.0001) while the unguided null keeps improving (Figure 2). Its gap against the null widens monotonically across the three matched epochs (Table 1).

We cannot read this as evidence that high coverage is harmful. An audit found that the collator shortens predictor targets *after* COVER chooses its placement, so its delivered masks hide *less* anatomy than ENVELOPE's (Table 2, Figure 3) and the arm never realised the coverage it was configured for. Appendix G gives the full account.

Table 2: Measured mask geometry, 600 training slices, production samplers. “purity” is the fraction of masked cells lying on tissue; “loss slots” counts supervised target-token instances, summed over the $M$ targets and including duplicates. AUC is quoted at the *matched* epoch 50, so the geometry-to-AUC association is not read across mixed epochs.

| policy | anatomy hidden | purity | mask ratio | context kept | loss slots | AUC @ep50 |
|---|---|---|---|---|---|---|
| RANDOM | 54.0% | 31.5% | 44.5% | 41.9% | 159.9 | 0.8641 |
| CENTROID | 62.1% | 40.0% | 40.3% | 45.5% | 159.0 | 0.8740 |
| ENVELOPE | 77.6% | 43.3% | 46.5% | 40.6% | 159.7 | **0.8761** |
| COVER | 73.5% | 44.2% | 43.2% | 43.2% | 159.1 | 0.8643 |
| ANATOMY-V2 | 79.9% | **97.1%** | 21.3% | 67.7% | 64.0 | 0.8654 |

**The strongest policy consults no segmentation model.** CENTROID finds its band from a per-column intensity centroid: no model and no annotation. Its margin over the null is the largest at epoch 100 and does not decay with training, whereas ENVELOPE's does (Table 1), and it exceeds the segmenter-guided ENVELOPE arm at epoch 100 by $+0.0047$ (CI $[+0.0004, +0.0091]$), though their paired intervals span zero at epochs 50 and 75.

### 5.2 Delivered geometry limits the interpretation

**The highest epoch-100 endpoint hides the least anatomy.** Of the four guided arms the one that hides the most (masking almost nothing but tissue, at $97.1\%$ purity) does not separate from the null. Both readings come from running each production sampler on 600 training slices and measuring the realised masks (Table 2, Figures 9 and 10; provenance in Appendix D). What CENTROID does instead is leave more visible: it retains more context than any other rectangle arm, and the background stays maskable. Rank correlations over these arms are positive for both anatomy hidden and purity, but at $n$=4–5 arms they resolve nothing in either direction and we draw no inference from them. Historical fine-tuned attribution images are not used to explain this result, because their numerical inputs are not retained (Appendix K).

**The anatomy arms are confounded on several axes simultaneously.** ANATOMY-V2 retains more context still, but it also hides far less of the grid and collapses to 64 predictor loss slots against about 159 for every rectangle arm (Table 2, Figure 11), changing the prediction task. They and COVER also differ from ENVELOPE in guide provenance (Appendix Q). Any of these could account for its deficit without invoking target shape at all. **H3 is therefore not identified by this design**, and we do *not* claim that irregular target shape is harmful.

These are not consequences of target shape: the anatomy arms resample every target to a fixed $K$=16 cells, so $4 \times 16 = 64$ loss slots follows by construction, while the rectangle arms do not set $K$. The two families are collated differently, which weakens “everything else held fixed” for the anatomy-versus-rectangle comparison specifically. Rectangle-family comparisons share the collation rule but not necessarily its realized context or target budgets (Appendix G).

**Production-path replay.** On fixed Training crops in full batches, historical COVER loses tissue targets through truncation in 300 of 576 views and leaves 55 with no guide-positive encoder context. Among 573 valid guides, 358 miss the final occupancy-cell context criterion defined in Appendix H. Scoring the exact delivered target prefixes removes target-truncation losses but leaves 320 context-floor misses. A separately enabled guide-aware context selection satisfies the floor in these valid-guide cases. These measurements identify a mismatch between the intended and delivered task, not the cause of the historical AUC ranking. The corrected policies have not been pretrained; details and diagnostic controls are in Appendix H.

### 5.3 Background targets and redundant downstream features

Unguided masking often selects background targets. At epoch 100 the null's predictor beats a per-position, no-context reference by 0.680 on *background* targets, *above* its 0.633 on anatomy: predicting a background token takes more than knowing where it sits, even though such tokens start out nearly pure position ($90.8\%$ of across-position input variance there comes from `pos_embed`, against $40.8\%$ at tissue). Pretraining spends self-similarity changes during training: under ENVELOPE, the one arm measured, self-similarity falls from $0.784$ untrained to $0.346$. In a separate epoch-50 regional probe using 25 slices and 2,000/600/1,000 training/validation/test volumes, pooled background features are $95.2\%$ linearly reconstructible from pooled anatomy features. Residualised, background predicts glaucoma at test AUC $0.5515$ (barely above chance, though the interval excludes it), and appending it to anatomy *lowers* test AUC for the null arm under that protocol. These diagnostics show that predicting background embeddings is nontrivial and that their incremental classification signal can be small. They do not establish why the epoch-100 baseline is strong or whether background target selection causes a downstream gain. The proposed causal controls were not run (Appendix I).

### 5.4 A larger observed gap in low-label probe refits

The observed margin is larger with fewer labels: at $5\%$ of the labelled set ($n$=300) CENTROID leads the null by $+0.0496$, against $+0.0108$ at full supervision. This is descriptive evidence from 5 shared subset refits per low-label fraction, not a tested gap-by-label-fraction interaction. Full-supervision fits agree with the corresponding primary entries to within $0.0009$ across the four arms. Appendix J reports the curve, variation and protocol.

### 5.5 Exploratory subgroup analysis

Across seven stratifications and 23 probes spanning aggregate AUC $0.8483$ to $0.8855$, the lowest-AUC eligible group is unchanged for five attributes. The metadata join uses deterministic loader order and reproduces the saved labels; excluded probes remain excluded. These probes share a test set and include multiple checkpoints from only 7 branches, so their subgroup rankings are descriptive, not independent replications. The apparent checkpoint-level association between race gap and aggregate AUC does not persist under branch aggregation (Appendix E).

At epoch 100, CENTROID has positive point-estimate gains in all race and disease-severity strata shown, but the adjusted race intervals exclude zero only for the white group. Severity strata are compared with the shared negative pool; only the mild interval excludes zero after the declared family adjustment (Tables 7 and 8). These results do not show that one stratum benefits more than another. The paired difference between gains in the lowest- and highest-AUC race groups is $-0.00091$ (CI $[-0.03285,\ +0.03104]$); the corresponding gender contrast also includes zero.

No general equity improvement is established. Other attributes include exceptions to the positive point-estimate pattern, and a shared validation-selected threshold transfers unevenly across strata (Appendix L). The full subgroup and race-by-gender analyses are in Appendices E and E.1. Masking policy is not presented as a fairness intervention.

### 5.6 Controls: the effect is not a precision artefact

The eight recorded fp32 re-probes shift AUC by less than $2 \times 10^{-4}$, orders of magnitude below the effects in Table 1, so the differing autocast settings of Section 4 are not the effect and every contrast is matched on precision (Appendix P). The generating script, the epoch-100 CENTROID encoder, its head and its stored per-case predictions are retained (links withheld for anonymity). Applying the recovered original head to the retained fp32 test-feature cache reproduces the stored AUC to $9.8 \times 10^{-6}$. This is agreement between the head, cached features and saved scores, not a fresh end-to-end encoder rerun.

## 6 Discussion and limitations

**What the evidence supports.** The segmenter-guided continuation has higher observed AUC by $+0.0120$ at epoch 50, and the intensity-centroid policy by $+0.0109$ at epoch 100, relative to their matched uniform baselines. Paired intervals exclude zero at all three observed epochs. The latter policy requires no segmentation model. More anatomically specific implementations do not have a consistent advantage, but this is not a controlled test of target shape or coverage: delivered context, target area, loss weighting and guide provenance differ. Background diagnostics motivate considering the information in targets and context jointly, rather than treating all background as wasted supervision. Their downstream causal contribution is unresolved.

**What it does not support.** We cannot claim anatomy-shaped masking is *harmful*: at the matched epoch it sits $+0.0013$ from the null with an interval ($[-0.0055,\ +0.0082]$) that comfortably contains zero, so it is not resolved as worse. Nor is target *shape* identified as the operative variable (Table 2), nor is CENTROID's mechanism. Two explanations remain live: consistency of location (the encoder is repeatedly forced to represent the same region), and masking ratio or task difficulty. Distinguishing them requires an arm that randomises the band's vertical position while holding area and context fixed, which we did not run.

**One pretraining continuation per policy.** The study uses a shared encoder ancestry and matched nominal training and probe settings. The mask sampler is the intended difference, though it also moves delivered mask ratio, retained context and loss slots, and across families collation and guide provenance (Section 5.2). What it does not resample is pretraining stochasticity, so each arm is one continuation. The paired bootstrap bounds sampling error on a fixed test set, so the intervals support statements about *these* continuations rather than an expected *ranking* over retrainings, which pairwise significance on one test set does not establish [Bouthillier et al., 2021]. Probe-seed variance is outside the design too: an earlier multi-seed probe check is not reproducible from retained artifacts, so we state no bound on probe noise. A planned six-leg replication is paused, with no completed epoch-50 replication result. It would add two seeds each for RANDOM, ENVELOPE and CENTROID from the epoch-25 ancestor, under generated configurations differing in policy, seed and logging fields. Seeds do not randomise the distributed sampler's visitation order. The proposed analysis is recorded in Appendix B; neither its existence nor a partial continuation increases the number of completed runs in Table 1.

**Scope, coverage and an incomplete arm.** All results are glaucoma classification from FairVision OCT. Generality to other modalities, diseases, or segmentation-quality regimes is untested, and the anatomy arms depend on MIRAGE's prediction quality on this data, so a better segmenter might change the conclusion. Only $f{=}0.21$ was pretrained for COVER, so we report no dose-response over the visible-tissue floor and make no claim that any floor is optimal. That arm reaches epoch 100, but the collation defect above (Appendix G) means its trajectory is not evidence about aggressive coverage. Its epoch-50 value is retained as a matched-epoch comparison against every other arm. Finally, one dataset and one test split were reused throughout this programme's history, and policies, checkpoints and analyses were chosen after repeated inspection of that split. The intervals we report are therefore best read as descriptive rather than as confirmatory inference.

**Broader impact and ethics.** This study uses a public, de-identified retinal dataset under its release terms and involves no new recruitment or data collection, and no model reported here is clinically validated or intended for deployment (Appendix M).

# 7 Conclusion

Tissue-directed rectangular target placement improves frozen-probe AUC in the observed OCT continuations, including a simple segmentation-free centroid policy. The results do not establish an equivalence between guidance methods or an expected ranking across retrainings. More targeted masking is not a single intervention when it also changes the context and target tokens delivered to the model. The mixed anatomy-shaped results and the coverage implementation defect therefore motivate joint control of target selection, retained context and background supervision. Their causal contributions remain open. A corrected, budget-matched multi-run comparison is subsequent work, not a result implied by the present diagnostics.

# References


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# A All frozen probes

Table 3 lists every frozen MeanPool probe in the study. It has 37 rows: 31 valid probes, plus two probes excluded and four retracted for the reasons in Appendix F. Two further counts follow from those 31. Eight of them are fp32 re-probes of an encoder already probed at fp16, and the subgroup audit collapses each such pair to the single encoder it scores twice, leaving 23 distinct encoder–epoch units (Section 5.5); 19 of those carry a stored per-group race summary. The two race analyses therefore run on different subsets by construction: the gap-trend test in Table 6 uses all 23 probes, since every one carries a race *gap*, while the scatter in Figure 5 needs a per-group AUC for all three groups and so uses the 19 probes that have one. The four that do not are COVER at epochs 73, 75 and 100 and ANATOMY-V2 at epoch 75: each was probed after the per-group race artifact was generated, and that artifact was not recomputed, so their absence is one of artifact chronology and not a property of those runs, whose race gaps are present and do enter the 23-probe trend test.

Table 3: Every frozen probe in the study, including runs excluded from the analysis and the reasons. The probe protocol is otherwise the same for every row (FairVision test $N$=3000, seed 42, frozen encoder, MeanPool + LayerNorm + Linear) and differs only in the precision column. “Precision” is the probe-time numerical precision, inferred from the stored prediction dtype, and is distinct from the pretraining target precision discussed in Appendix F.

| policy | epoch | precision | test AUC | 95% CI |
|---|---|---|---|---|
| ANATOMY-V1 | 30 | fp32 | 0.8583 | [0.8450, 0.8712] |
| ANATOMY-V2 | 35 | fp32 | 0.8661 | [0.8531, 0.8787] |
| ANATOMY-V2 | 40 | fp32 | 0.8683 | [0.8553, 0.8807] |
| ANATOMY-V2 | 50 | fp32 | 0.8654 | [0.8522, 0.8781] |
| ANATOMY-V2 | 75 | fp32 | 0.8612 | [0.8477, 0.8740] |
| ancestor | 25 | fp32 | 0.8487 | [0.8349, 0.8621] |
| COVER | 27 | fp32 | 0.8483 | [0.8346, 0.8617] |
| COVER | 30 | fp32 | 0.8522 | [0.8386, 0.8654] |
| COVER | 34 | fp32 | 0.8571 | [0.8437, 0.8700] |
| COVER | 50 | fp32 | 0.8643 | [0.8513, 0.8768] |
| COVER | 73 | fp32 | 0.8647 | [0.8517, 0.8772] |
| COVER | 75 | fp32 | 0.8639 | [0.8507, 0.8763] |
| COVER | 100 | fp32 | 0.8577 | [0.8443, 0.8707] |
| ENVELOPE | 30 | fp32 | 0.8539 | [0.8405, 0.8669] |
| ENVELOPE | 50 | fp16 | 0.8761 | [0.8635, 0.8879] |
| ENVELOPE | 50 | fp32 | 0.8761 | [0.8635, 0.8879] |
| ENVELOPE | 75 | fp16 | 0.8803 | [0.8677, 0.8922] |
| ENVELOPE | 75 | fp32 | 0.8803 | [0.8677, 0.8922] |
| ENVELOPE | 100 | fp16 | 0.8807 | [0.8683, 0.8926] |
| ENVELOPE | 100 | fp32 | 0.8808 | [0.8683, 0.8927] |
| CENTROID | 50 | fp16 | 0.8740 | [0.8615, 0.8862] |
| CENTROID | 50 | fp32 | 0.8740 | [0.8614, 0.8862] |
| CENTROID | 75 | fp16 | 0.8836 | [0.8714, 0.8955] |
| CENTROID | 100 | fp16 | 0.8855 | [0.8735, 0.8971] |
| CENTROID | 100 | fp32 | 0.8853 | [0.8732, 0.8971] |
| RANDOM | 50 | fp16 | 0.8641 | [0.8510, 0.8769] |
| RANDOM | 50 | fp32 | 0.8641 | [0.8511, 0.8769] |
| RANDOM | 75 | fp16 | 0.8723 | [0.8593, 0.8849] |
| RANDOM | 75 | fp32 | 0.8723 | [0.8593, 0.8849] |
| RANDOM | 100 | fp16 | 0.8746 | [0.8618, 0.8869] |
| RANDOM | 100 | fp32 | 0.8745 | [0.8617, 0.8868] |
| anatomy-v2 | 75 | fp32 | 0.8625 | *excluded* |
| anatomy-v2 | 92 | fp32 | 0.8604 | *excluded* |
| random | 30 | fp32 | 0.8558 | *retracted* |
| random | 50 | fp32 | 0.8590 | *retracted* |
| random | 75 | fp32 | 0.8612 | *retracted* |
| random | 100 | fp32 | 0.8607 | *retracted* |

# B Continuation-level replication

**Status at submission: paused; no completed replication result.** The first leg has a completed epoch-26 checkpoint and an incomplete epoch 27, not an epoch-50 probe. What follows is the recorded design for subsequent work, not additional evidence for the present comparisons.

**Design.** Six continuations (RANDOM, ENVELOPE and CENTROID, each at two further seeds) resume from the same epoch-25 ancestor used by every arm in this paper. The ancestor is locked by content: SHA-256 `e5ad5b0c2aadfa15449409786afbfa39d8b5405b699be8f02f2e540195e97e7b`, 1,507,519,602 bytes, re-hashed immediately before each leg, and a mismatch aborts the leg rather than continuing from an unverified starting point. The six configurations are generated by a script that asserts they are identical outside the mask curriculum except for the seed and logging fields. Each leg runs to epoch 50 (the endpoint at which every arm already carries a frozen probe) with the optimiser, cosine learning-rate, weight-decay and EMA schedules still sized for 100 epochs, so the trajectory is the one this paper measures and not a shorter, differently-scheduled substitute. Each completed leg is then probed under the frozen-probe protocol of Section 4, with the encoder hashed before and after the probe. Together with the continuations reported here, this yields three continuations per policy at a matched endpoint.

**Recorded analysis plan.** (i) All nine epoch-50 continuation AUCs would be reported in one table, whatever they show. No continuation is dropped for any reason other than a documented crash, and a crash is reported. (ii) We report the per-policy mean and full range, and the two paired-by-seed differences ENVELOPE−RANDOM and CENTROID−RANDOM, one per seed. (iii) At three continuations we report point estimates and ranges only: no $t$-test, no $p$-value and no confidence interval at the continuation level, because a small set of continuations gives a limited account of training variability. (iv) The fixed-data-order caveat below is restated wherever the result is.

**What the seed does and does not randomise.** The seed randomises random-resized-crop draws, target and context block sampling (including the ENVELOPE rejection sampler and the CENTROID band sampler), dropout, and worker streams. It does *not* randomise the order in which slices are visited: the distributed sampler is constructed with its own fixed seed and advanced per epoch, which is true of the originally reported continuations as well. The unit of inference is therefore post-fork stochastic optimisation noise at a fixed data order, which is narrower than independent retraining, and it is not a replication of the data pipeline.

**The failure case, written in advance.** If the epoch-50 ordering does not reproduce (a paired difference changes sign, or the three-continuation range for either contrast spans zero), the reported conclusion becomes that the single-continuation contrasts of Table 1 describe those particular runs and not the policies, and every policy-level statement in this paper is downgraded to a run-level one. Table 1 is not withdrawn in that case: it remains a correct description of the runs it measures. That outcome is an anticipated and informative result of this design rather than a failure of it, which is why the rules above are fixed now.

**What this replication does not do.** It does not supply an untouched evaluation cohort. The comparison still runs on the same test split, which has been inspected repeatedly across this programme, so the replication bounds run-to-run variability and not adaptive selection. The number of times that split was inspected while choosing policies, checkpoints and analyses was not logged and cannot now be reconstructed, which is why every interval in this paper is described as descriptive rather than confirmatory. An untouched evaluation would require a labelled cohort from a different scanner or population, thresholds and checkpoints selected on that cohort's own validation split, a single evaluation, and no iteration afterwards. No such cohort was available to us.

# C Masking policies across the volume

Figure 4 compares final target and context sets on one predeclared view. It illustrates the implementations, not population stability across depth. The tissue reference and colour assignment are shared.

# D Mask-geometry provenance

Table 2 was regenerated from the production samplers rather than carried forward from an earlier draft, and this appendix records how, together with two facts the table itself cannot show: how much of each cell is measurement noise, and how one configuration constant moves the correlation quoted in Section 5.2.

**How the geometry was measured.** Each production sampler was run on the same 600 FairVision *Training* slices (24 volumes, 25 slices each), one image at a time, with the COVER visible-tissue floor at its production value $f$=0.21, on the $16 \times 16$ patch grid, with anatomy taken from the MIRAGE guide occupancy channel at threshold 0.25. The measurement was repeated with three independent draws (seeds 42, 1234 and 2026). Training slices are not a choice: the cached anatomy guides exist for the Training split only, so three of the five production samplers cannot be run on held-out slices at all.

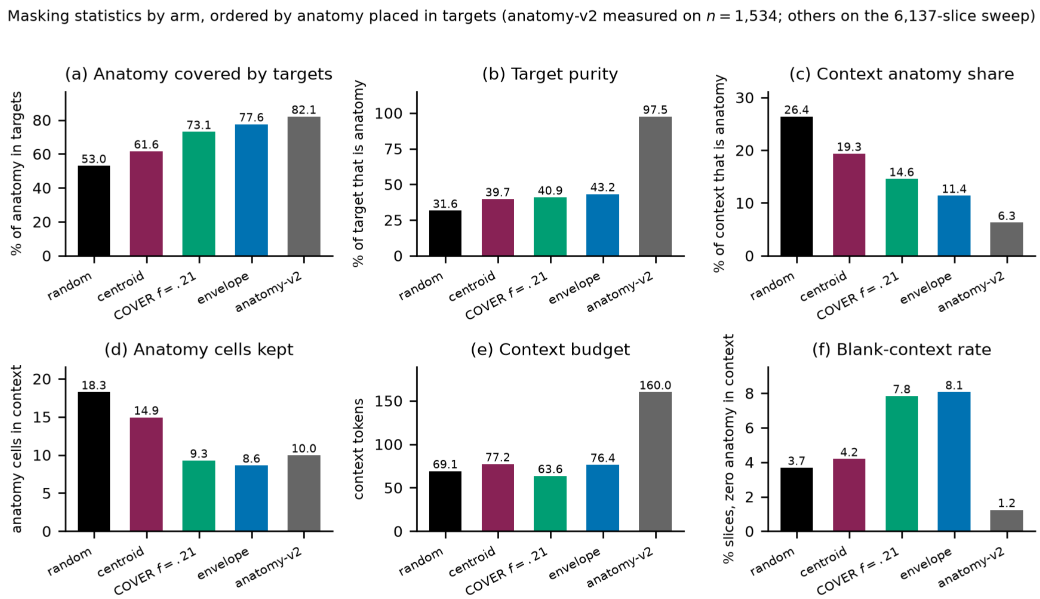


Figure 3: What each sampler actually does, measured on the 6,137-slice sweep (ANATOMY-V2 on $n$=1,534). Panel (a) is the quantity the COVER arm was built to maximise: it reaches 73.1%, *below* ENVELOPE's 77.6%, which is the discrepancy Appendix G explains. Panel (b) shows the ordering this paper tests: target purity rises monotonically from RANDOM to ANATOMY-V2, while downstream AUC does not follow it. Panels (c)–(f) give the confounds we do not disentangle: how much anatomy survives in the context (c, d), the context budget (e), and how often a slice is left with no anatomy visible at all (f). Every bar starts at zero and is a mean over the sweep, with no interval drawn; panel scales differ, so bars should be compared only within a panel.

**Cell-level agreement and seed noise.** Table 2 uses the seed-42 draw. Table 4 reports variability over the three measurement draws, not variability across pretrained encoders. No conclusion depends on the ordering of the small differences in mean rectangle-arm loss-slot counts. The rank order of the arms on anatomy hidden and on purity is identical across all three draws, which is what the rank correlations in the main text rely on.

Table 4: Stability of two Table 2 columns across three independent draws of the same 600-slice measurement. "sd" is the standard deviation over seeds 42, 1234 and 2026. These SDs quantify sampler redraw variation, not uncertainty over independently pretrained encoders.

| | anatomy hidden (%) | | loss slots | |
|---|---|---|---|---|
| policy | value | sd | value | sd |
| RANDOM | 53.95 | 0.71 | 159.91 | 1.15 |
| CENTROID | 62.13 | 0.64 | 158.99 | 0.43 |
| ENVELOPE | 77.58 | 0.38 | 159.68 | 1.00 |
| COVER | 73.55 | 0.30 | 159.09 | 0.45 |
| ANATOMY-V2 | 79.89 | 0.39 | 64.00 | 0.00 |

**The delivered context is not the per-image context.** The "context kept" column of Table 2 is per-image geometry, measured before collation. In training, every context mask in a batch is truncated to the batch minimum, so what the encoder actually receives at the production batch size is smaller, and unevenly so (Table 5). The context confound between the anatomy arm and the rectangle arms that Section 5.2 declines to explain away is consequently *larger* than the printed table suggests: roughly a doubling rather than a $1.6\times$ ratio. This strengthens the verdict that H3 is not identified by this design.

Table 5: Visible context as a fraction of the patch grid, same 600 slices and same seed, measured per image and as delivered to the encoder at the production batch size of 64. The per-image column reproduces Table 2's "context kept" to within $0.4$ points.

| policy | per image | delivered at batch 64 |
|---|---|---|
| RANDOM | 41.9% | 24.7% |
| CENTROID | 45.5% | 32.9% |
| ENVELOPE | 40.6% | 30.7% |
| COVER | 43.2% | 30.1% |
| ANATOMY-V2 | 67.7% | 63.5% |

**The rank correlation depends on a configuration constant.** The $+0.80$ anatomy-hidden/AUC Spearman coefficient of Section 5.2 is computed at the production COVER floor $f{=}0.21$. Re-measured on the same slices with the floor at $0.15$, COVER hides $79.5\%$ of anatomy instead of $73.4\%$ at the same batch size and overtakes ENVELOPE, which reorders the four rectangle arms and moves the same coefficient to $+0.40$. A rank statistic over four arms that a single configuration constant can move by that much is not evidence about anatomy, in either direction, which is why Section 5.2 draws its conclusion from the direct comparison instead.

**What would identify target shape.** An arm that isolated target shape would have to match ENVELOPE on the axes that currently differ, and those requirements are numeric, not rhetorical. ANATOMY-V2 sets four predictor targets resampled to a fixed $K{=}16$ cells, giving exactly 64 loss slots. Matching the rectangle arms' 158–160 slots at four targets requires $K \approx 40$ (inferred arithmetic on those measured values). It would also have to deliver a mask ratio of $40$–$46\%$ of the grid rather than the $21\%$ that the anatomy arms deliver, roughly twice the target area, which for a shape-following sampler means more or larger connected components and therefore changes the very shape statistics being tested, and it would have to pass through the rectangle collation path, including the batch-minimum truncation above. It would then need the same replicated continuation design and the same endpoint as Appendix B. We did not run it. Naming its specification is what we can offer instead.

## E Subgroup and severity analysis

All numbers below come from saved test predictions joined to FairVision's released metadata in deterministic loader order. The join is validated by exact label reconstruction (3000/3000 labels recovered in every probe). The 23 probes used here are exactly those the inventory marks valid: the four retracted RANDOM probes of Appendix F and the two precision-spliced ANATOMY-V2 probes are excluded here as everywhere else in the paper.

**Cohort composition and analysis eligibility.** The composition counts below describe the complete test cohort, without imputation. They are not the eligibility rules for every subgroup analysis. The released metadata carries every attribute we use for all 3000 test subjects, whose identifiers are all distinct, and the label is complete at 1466 positive and 1534 negative. Race is 251 asian, 431 black and 2318 white; gender is 1,716 female and 1,284 male; ethnicity is 2,894 non-hispanic and 106 hispanic; language is 2,782 english, 174 other and 44 spanish; marital status is 1,741 married or partnered, 776 single, 199 widowed, 191 divorced, 71 unknown and 22 legally separated; age is present and numeric for all 3000, spanning $9.5$–$98.0$ years. Each of those five categorical breakdowns sums to 3000 exactly. Analysis-specific exclusions are described below.

One level is non-substantive rather than absent: 71 of 3000 subjects ($2.4\%$) carry marital status *unknown* in the release. It is included in these composition counts but omitted from the seven-attribute trend analysis, not folded into another group. Severity is derived from visual-field mean deviation, and its 334 severe, 460 moderate and 672 mild strata sum to all 1466 positives, so no positive is lost at a stratum boundary. The release uses a mean-deviation sentinel of $-1$ in 3 of the 3000 test records. All 3 carry a negative label, and because every severity stratum is scored against the undifferentiated pool of all 1534 negatives, that sentinel never enters a stratum definition.

The per-group fairness-summary workflow marks groups with fewer than 50 cases or fewer than 10 of either class as underpowered and omits them from its max–min summaries. The separate seven-attribute trend workflow instead uses a minimum of 40 cases and skips unknown, blank and `na` levels. Consequently, Spanish is included there, whereas unknown marital status and the undersized legally separated group are omitted. The eligibility rules are not interchangeable, and no statistics were regenerated under a different rule to make the workflows agree.

**Why we report two sample sizes.** The 23 probes span only 7 pretraining branches, because several are different epochs of one training run. Probes within a branch share an encoder lineage, a seed and the test split, so treating them as 23 independent observations is pseudo-replication. Table 6 therefore reports each trend twice: once per checkpoint, and once after averaging within each branch. Where the two disagree, the branch-level figure is the one to believe.

Table 6: Subgroup gap trends. "gap range" is the max–min subgroup AUC gap observed across the 23 valid probes. $q$ is Benjamini–Hochberg across the seven checkpoint-level tests. Branch $p$-values are unadjusted across the same seven and, at $n$=7, descriptive only. Only gender survives checkpoint-level correction at the conventional 0.05 level, and it *narrows* as aggregate AUC rises. Race, ethnicity and disease severity share the same $q$ just above that level, so no threshold admits race and severity without also admitting ethnicity. Race is the one attribute whose checkpoint-level trend does not persist under branch aggregation ($p$=0.4821), where the effective sample is 7 independent branches rather than 23 probes; we therefore draw no trend conclusion for race.

| | | | per checkpoint ($n$=23) | | | per branch ($n$=7) | |
|---|---|---|---|---|---|---|---|
| attribute | worst group | gap range | $\rho$ | $p$ | $q$ | $\rho$ | $p$ |
| sex | female (23/23) | 0.0272–0.0407 | -0.664 | 0.0005 | 0.0038 | -0.821 | 0.0234 |
| race | black (23/23) | 0.0391–0.0935 | +0.473 | 0.0225 | 0.0668 | +0.321 | 0.4821 |
| ethnicity | hispanic (23/23) | 0.0180–0.0661 | +0.435 | 0.0381 | 0.0668 | +0.679 | 0.0938 |
| language | other languages (19/23) | 0.0518–0.0805 | +0.311 | 0.1483 | 0.1730 | +0.786 | 0.0362 |
| marital status | single (23/23) | 0.0286–0.0654 | -0.210 | 0.3351 | 0.3351 | -0.679 | 0.0938 |
| age | $<$60 (22/23) | 0.0430–0.0704 | +0.399 | 0.0591 | 0.0828 | +0.357 | 0.4316 |
| disease severity | mild $(-6,-2]$ (23/23) | 0.1286–0.1400 | -0.449 | 0.0318 | 0.0668 | -0.821 | 0.0234 |

**Race.** The black subgroup has the lowest AUC in every one of the 23 probes ($n$=431 black, $n$=251 asian, $n$=2318 white; Figure 5). At matched epoch 100 the black subgroup improves from 0.8325 under RANDOM to 0.8472 under CENTROID (+0.0147), a larger absolute gain than the white subgroup's ($0.8741 \rightarrow 0.8853$). The max–min gap nonetheless widens slightly, because the asian subgroup (already the highest) gains most ($0.9033 \rightarrow 0.9189$). A widening max–min gap with universally rising subgroup AUCs is not evidence of harm. The checkpoint-level correlation between aggregate AUC and racial gap is $\rho = +0.473$ ($p$=0.0225). It does not pass Benjamini–Hochberg correction at the stated conventional threshold ($q$=0.0668), and it vanishes once probes are aggregated to their pretraining branches ($\rho = +0.321$, $p$=0.4821), and the checkpoint-level test is pseudo-replicated because the 23 probes come from only 7 branches.

**Disease severity.** FairVision's label is defined by visual-field mean deviation (md): every test volume with md $\leq -2$ is positive. Severity therefore cannot be used as an ordinary subgroup axis, because within-bin AUC is undefined, a trap that silently yields nothing under naive stratification. Instead each positive stratum is scored against the shared pool of all 1534 negatives (Table 7).

Table 7: Severity-stratified AUC at matched epoch 100. Each positive stratum is scored against the shared pool of all 1534 negatives. Each point estimate rises; the gains were not tested against one another.

| stratum | $n_+$ | RANDOM | ENVELOPE | CENTROID | $\Delta$ |
|---|---|---|---|---|---|
| severe (md $\leq -12$) | 334 | 0.9525 | 0.9559 | 0.9588 | +0.0063 |
| moderate ($-12 <$ md $\leq -6$) | 460 | 0.9030 | 0.9103 | 0.9131 | +0.0102 |
| mild ($-6 <$ md $\leq -2$) | 672 | 0.8164 | 0.8232 | 0.8301 | **+0.0137** |

The severe-to-mild gap is 0.1361 for the null and 0.1286 for the strongest policy, and stays within 0.0114 (0.1286–0.1400) across all 23 probes while aggregate AUC spans 0.8483–0.8855. All three point estimates rise, but family-wise simultaneous adjustment leaves only mild excluding zero. Moderate and severe no longer do (Table 8). The largest point estimate is at mild disease ($+0.01372$, CI $[+0.00204,\ +0.02541]$) and the smallest at severe ($+0.00626$). That ordering is descriptive only: we did not test the paired *difference* between strata, so it does not establish that mild disease benefits most. Mild functional loss is harder to separate from normal OCT by construction, so the difficulty ordering is expected clinically. What the data add is that no masking policy we tested changes it.

Table 8: Paired change in subgroup AUC from RANDOM to CENTROID at matched epoch 100. The eight rows plus two gain-difference contrasts in the text form one exploratory family. Reported CIs are 95% single-step max-$|t|$ simultaneous intervals from 10,000 outcome-stratified paired subject resamples shared across both fixed heads and every stratum (family-wise $\alpha = 0.05$). Four of ten exclude zero: mild severity, white race, and both gender strata.

| stratum | $n$ | $\Delta$ AUC | simultaneous 95% CI | excludes 0 |
|---|---|---|---|---|
| severity, mild | 2206 | +0.01372 | [+0.00204, +0.02541] | yes |
| severity, moderate | 1994 | +0.01016 | [-0.00029, +0.02061] | no |
| severity, severe | 1868 | +0.00626 | [-0.00135, +0.01386] | no |
| race, White | 2318 | +0.01129 | [+0.00286, +0.01972] | yes |
| race, Black | 431 | +0.01467 | [-0.00742, +0.03676] | no |
| race, Asian | 251 | +0.01558 | [-0.00766, +0.03882] | no |
| sex, Female | 1716 | +0.01152 | [+0.00100, +0.02204] | yes |
| sex, Male | 1284 | +0.01006 | [+0.00067, +0.01945] | yes |

**Limitations.** The $p$-values here are descriptive, for the reasons given in Section 6. Race and ethnicity are self-reported categories inherited from FairVision's coding. A race-by-gender intersectional breakdown *is* reported, in Appendix E.1. Its limits are that two of its six cells hold fewer than 130 cases and that it carries no paired per-cell intervals, so it describes the disparity and tests no per-cell change. All figures are frozen mean-pool linear probes. Subgroup sensitivity, specificity and race-stratified expected calibration error at a transferred threshold are reported in Appendix L. What is absent is subgroup predictive values, reliability curves, and calibration stratified by any attribute other than race, so this is not a complete fairness evaluation. Disparities could differ under fine-tuning or a stronger head.

### E.1 Intersectional breakdown: race by gender

The audit above crosses no attributes, so on its own it cannot say whether a group disadvantaged on two axes at once is worse served than either margin implies. We therefore also computed AUC for all six race-by-gender cells of the test split, from the same saved predictions, under the same exact-label join proof, with per-arm, 2,000-resample class-stratified percentile intervals within each cell and fitted heads fixed. The second axis is FairVision's self-reported gender coding, which we retain here without reinterpreting it as a biological-sex measure. Legacy figures using "sex" refer to this same released variable. The artifact behind this subsection covers 22 arms, 18 of them non-retracted. Those 18 are a strict subset of the 23 probes of Table 6. Five are absent: COVER at epochs 50, 73, 75 and 100, and the ANATOMY-V2 epoch-75 probe, whose run directory carries the internal name `blob`, an artifact label for the anatomy-shaped family that this paper calls ANATOMY-V2, and the only place that label survives here. All five were probed after this artifact was generated and it was not recomputed, so their absence is artifact chronology rather than an analytic exclusion. The four retracted arms the artifact does contain are dropped for the reasons in Appendix F. Every count in this subsection is therefore out of 18 arms, not 23.

**The ordering compounds, and it never reorders.** The black $\times$ female cell has the lowest AUC in all 18 non-retracted arms and the asian $\times$ male cell the highest in all 18. Within each race the female cell sits below the male cell in all $54$ race-by-arm combinations, and within each gender the black cell sits below the white cell in all 36. These 18 arms share one test split, one epoch-25 ancestor and one probe seed, and several are different epochs of a single branch, so this unanimity is a consistency check across checkpoints, not 18 replications. We attach no $p$-value to it, exactly as for the marginal ordering above.

Table 9: Race-by-gender AUC at matched epoch 100 on the 3000 test volumes, with per-cell counts and per-arm bootstrap percentile intervals. $\Delta$ is CENTROID minus RANDOM as a difference of point estimates. The artifact stores per-arm intervals only and no paired per-cell resampling, so no $\Delta$ in this table is tested against zero; per-cell significance is not reported. The 18 intervals are marginal, not simultaneous. The two asian cells are small ($n$=123 and $n$=128) and their intervals are correspondingly wide.

| cell | $n$ | $n_+$ | RANDOM | ENVELOPE | CENTROID | $\Delta$ |
|---|---|---|---|---|---|---|
| asian × female | 123 | 52 | 0.8761 [0.8024, 0.9342] | 0.8762 [0.8042, 0.9366] | 0.8904 [0.8245, 0.9423] | $+0.0144$ |
| asian × male | 128 | 66 | 0.9245 [0.8752, 0.9655] | 0.9145 [0.8565, 0.9626] | 0.9409 [0.8983, 0.9751] | $+0.0164$ |
| black × female | 250 | 149 | 0.8195 [0.7668, 0.8679] | 0.8155 [0.7632, 0.8638] | 0.8309 [0.7790, 0.8783] | $+0.0114$ |
| black × male | 181 | 119 | 0.8435 [0.7822, 0.8994] | 0.8652 [0.8091, 0.9141] | 0.8663 [0.8132, 0.9149] | $+0.0228$ |
| white × female | 1343 | 610 | 0.8641 [0.8441, 0.8826] | 0.8728 [0.8530, 0.8910] | 0.8762 [0.8570, 0.8943] | $+0.0121$ |
| white × male | 975 | 470 | 0.8864 [0.8650, 0.9062] | 0.8958 [0.8744, 0.9148] | 0.8962 [0.8746, 0.9153] | $+0.0099$ |
| max−min gap | | | 0.1050 | 0.0990 | 0.1100 | |

**The marginal race gap understates the worst cell.** Averaged over the 18 arms, the max−min gap is $0.0340$ by gender alone, $0.0653$ by race alone, and $0.1046$ across the six race-by-gender cells, and the intersectional gap exceeds that arm's marginal race gap in all 18. The race margin therefore understates the spread a race-by-gender reading exposes by $60.1\%$ on average, arithmetic on those two means, not a separate measurement. The same pattern holds at the matched epoch of Table 9: under CENTROID the marginal race gap is $0.0717$ while the race-by-gender gap is $0.1100$, and the worst cell, black × female at $0.8309$, sits $0.0163$ below the marginal black figure $0.8472$ that the marginal audit above reports. An audit that stops at the margins reports a worst-group number that is optimistic relative to the worst cell. The compounding is close to additive and we do not claim more: the gender and race means sum to $0.0993$ against an observed $0.1046$, a ratio of $1.053$, which is not evidence of a super-additive interaction.

**Not every cell improves.** The statement elsewhere in this appendix that every subgroup point estimate rises from RANDOM to CENTROID is a statement about marginal strata at the matched epoch, and it does *not* extend to cells at every epoch. At epoch 100 all six point estimates rise (Table 9), but at epoch 50 the asian × female cell falls by 0.0020, and ENVELOPE minus RANDOM is negative in three of six cells at epoch 75 (asian × female $-0.0148$, asian × male $-0.0031$, black × female $-0.0025$) and in two of six at epoch 100 (asian × male $-0.0100$, black × female $-0.0040$, with asian × female at $+0.0001$). No policy we tested raises every intersectional cell at every epoch, and the exceptions fall in the small asian cells and in the worst cell.

**What this does not establish.** There are no paired per-cell bootstrap intervals in the artifact, so every $\Delta$ above is a difference of point estimates and none of them is separated from zero. Per-cell delta significance is not computed here, though it is computable from the retained per-arm predictions. Two cells carry $n$=123 and $n$=128, so their intervals overlap almost everything. Cells with a single class or fewer than 40 cases would have been dropped. None were, so all six race-by-gender cells are reported. Nothing here is a fairness intervention: as with the margins, no policy reorders the cells, and the worst-served cell is the same under the unguided null and under every guided policy we tested.

## F Excluded runs

Four probes of an earlier *null* run are excluded from all analysis. They are unguided-masking controls from a superseded coverage campaign: the inventory lists them under RANDOM, and their run tags begin `frozen_cover_random` because of that campaign, not because they use a coverage policy. They were trained with half-precision EMA targets and a different encoder-truncation policy from the arms they would be compared against, so their deficit is not attributable to masking. Similarly, the ANATOMY-V2 probes at epochs 75 and 92 follow a change of target precision partway through that arm's training. They are listed in Table 3 but excluded from the matched-epoch comparisons in Table 1.

## G A collation defect in the coverage arm, and what it invalidates

We audited the COVER sampler after its epoch-100 result and found a defect. It invalidates the reading of that arm and weakens one cross-family comparison.

**The defect.** Predictor targets are collated by truncating every target to the length of the shortest target anywhere in the microbatch. COVER chooses its placement *before* this happens: it greedily maximises hidden anatomy subject to a visible-tissue floor, evaluated on full rectangles, and the collator then shortens those rectangles. The optimisation is therefore defeated after the fact.

**Measured consequences.** A one-off audit found that collation reduced COVER's anatomy coverage. An independent sweep over $6{,}137$ slices gives $73.1\%$ against $77.6\%$ for ENVELOPE. The arm intended to hide *more* anatomy than ENVELOPE in fact hides *less*. Across $24{,}000$ emitted targets only $73.4\%$ remain perfect rectangles. Because target cells are enumerated row-major, truncation keeps the top of each rectangle and discards the bottom, so the loss is directional rather than a neutral shrink. *Provenance:* the initial one-off CPU audit was not persisted, so neither its sample count nor its pre/post-truncation values are used as quantitative evidence. The older $73.1\%$ against $77.6\%$ comparison is backed by the stored floor sweep. The new source-bound production-path replay is reported separately in Appendix H.

**Why it went unnoticed.** Realised coverage is recorded before truncation. Every training log therefore reported the intended figure, and the instrument agreed with the design rather than with the data.

**What it invalidates.** We cannot claim that COVER tests aggressive anatomy coverage, nor that its decline shows over-covering anatomy is harmful. On delivered masks it does not out-cover ENVELOPE. Its measured trajectory remains a real property of the policy as implemented. The mask geometry in Table 2 is unaffected, having been measured on delivered masks.

**Scope.** ENVELOPE shares the same collation path and the same directional clipping, but sets no exact coverage target, so it has no comparable optimisation to defeat. The anatomy arms instead resample every target to a fixed $K{=}16$ cells. Contrasts within the rectangle family are collated identically and are unaffected. The anatomy-versus-rectangle contrast is not, and we treat it accordingly in Section 5. A corrected coverage arm must be retrained from the shared ancestor, since patching mid-trajectory would splice two masking policies into one curve.

**Status of a corrected run.** At submission the corrected arm is not run. Until it exists, the question this arm was built to answer (whether hiding most of the anatomy is harmful) remains *open*.

## H Delivered-task replay and versioned corrections

**Scope.** We replayed the actual guided dataset, paired transforms, DataLoader worker collator and returned mask tensors on the existing 600-view Training scope (24 volumes, 25 views each), using fixed new crop draws shared across policies. These are not reconstructed historical training crops. Drawn block sizes were injected independently of placement; equal seeds alone do not pair policy-dependent rejection sampling. The production-size subset contains 576 views in complete batches of 64, with 573 valid post-crop guides. The shorter final batch is excluded from Table 10. Views within volumes are correlated, so these are descriptive engineering counts, not independent training replications.

**What is measured.** Target coverage is measured both at scoring and after target collation. Encoder-visible tissue is counted only in the final encoder indices, not in the full-grid target complement. Tissue is a guide occupancy thresholded at $0.25$, not clinical ground truth. The context criterion is at least $\max(\lceil 0.21 N_{\text{tissue}} \rceil, \min(4, N_{\text{tissue}}))$ non-target tissue cells. It is an explicit occupancy-cell diagnostic and corrected-policy guarantee, distinct from the historical soft-mass constraint on the target complement.

Table 10: Full-batch delivered-task replay on 576 Training views. Floor misses count only the 573 valid guides; invalid guides are never certified as satisfying a guarantee. Scored and delivered coverage refer to supported soft guide mass. Context counts include all views. The corrected columns are sampler diagnostics, not pretrained-model results.

| Measurement | Historical COVER | Exact-prefix v2 | v2 + context guard |
|---|---|---|---|
| Views losing tissue targets | 300 | 0 | 0 |
| Valid-guide context-floor misses | 358 | 320 | 0 |
| Views with zero tissue context | 55 | 38 | 1 |
| Scored hidden guide mass (%) | 78.51 | 78.56 | 78.56 |
| Delivered hidden guide mass (%) | 74.57 | 78.56 | 78.56 |
| Mean encoder-visible tissue cells | 10.13 | 11.24 | 15.12 |

**Two separate corrections.** The opt-in exact-prefix implementation scores the target shapes that will actually be returned, preserving the drawn rectangle sizes and delivered target-token budget. Context exclusion then uses delivered targets rather than also excluding discarded candidate cells. The optional context guard reselects non-target tissue while retaining the final context token count. It can select outside the original context rectangle and is therefore a second guide-aware intervention, not a pure placement correction. Targets and token budgets are exactly matched between v2 and guarded v2. The one zero-tissue context in the guarded column has an invalid guide and is explicitly uncertified. Historical behavior remains separately replayable.

**Random fill is not a background quota.** At full ramp all usable targets take the guided branch, while the `random_legal` fill mechanism is conditional on unused target blocks. Some views have no random-fill blocks. Such blocks can include tissue, and guided blocks can include background, so branch labels cannot stand in for background supervision. Corrected target scoring does not introduce a guaranteed random/background quota. Whether another mixture improves representations requires a separately defined comparison.

**The delivered tensors reach the objective.** Using exact masks for one fixed batch of the first 2 predeclared views, we ran 3 independent fp32 one-update checks from the same historical ancestor, one per COVER implementation. Tissue/background and first/repeated token contributions reconcile with the actual Smooth-L1 loss. Online/predictor gradients and parameter updates were finite and nonzero; teacher gradients were absent and EMA updates matched their stated formula. The guard changed 0 contexts in these selected views, which already met its floor; equality with unguarded v2 is therefore expected. These bounded checks neither train a corrected representation nor estimate its downstream AUC.

**Other training safeguards.** The implementation now normalizes short final accumulation windows by their actual length and advances schedules and EMA only after successful optimizer updates. New epoch-boundary checkpoints retain continuation state with explicit exact-resume and intentional-fork modes. CPU regression witnesses establish their tested contracts; they do not reconstruct historical overflow counts, persistent-worker streams or multi-rank execution. No existing checkpoint or prediction has been reassigned to corrected code. A corrected multi-arm pretraining comparison remains subsequent work.

# I Background representation diagnostics

Unguided masking spends most of its predictor targets on background, yet it reaches within $+0.0109$ of the best policy. We investigated whether background carries diagnostic signal, and whether position embeddings are the mechanism.

**Background is a real *pretraining* signal.** At epoch 100 the random arm's predictor beats a per-position, no-context reference by $0.680$ on background targets, above its $0.633$ on anatomy targets: predicting a background token requires more than knowing where it is. Pretraining also spends capacity reorganising background in the separate ENVELOPE analysis: background self-similarity falls from $0.784$ untrained to $0.346$ at epoch 100.

**Incremental signal under a regional-probe protocol.** This analysis uses epoch-50 encoders, 25 slices per volume and 2,000/600/1,000 training/validation/test volumes, rather than the main 100-slice protocol and full split. Pooled background features are $95.2\%$ linearly reconstructible from pooled anatomy features. On $n{=}1000$ test subjects, with fitted transforms and probes fixed, residual background has AUC $0.5515$ (2,000-resample case-percentile CI $[0.5165, 0.5893]$). A paired bootstrap of the same size puts the change from appending background at $-0.0076$ (CI $[-0.0139, -0.0012]$) for the random arm and under $\pm 0.002$ for every other. A background-only probe scores $0.867$. Self-attention mixes tissue information into background tokens, so that figure cannot be read as background signal, and we did not measure how much mixing occurs.

**Position dominates background tokens, but is not the whole story.** At layer $0$, $90.8\%$ of the across-position variance of the encoder input at background cells comes from `pos_embed`, against $40.8\%$ at tissue cells (stable across six thresholds and both checkpoints). Mechanically a background token is close to its position code. That the predictor still beats a per-position reference shows the pretraining signal is not purely positional. The causal chain from this mechanism to downstream AUC remains open.

**The two controls that would settle it, neither of which was run.** First, a regional probe on an *eroded* background region (background cells at a fixed minimum distance from any tissue cell), which removes the tokens most exposed to local mixing while keeping the probe protocol fixed. Second, a background-content shuffle in which background token *contents* are permuted across images while positions and token count are held fixed: if the background-only AUC survives that, the signal is positional rather than content-bearing. Neither has been run, so this appendix reports a mechanism that is described but not established.

## J Label efficiency

We measured each policy as supervision is withdrawn (Section 5.4). We subsample the labelled training set to fractions of its size, fit the same frozen linear probe on cached epoch-100 features, and repeat 5 times per fraction. The per-fraction seed depends only on fraction and repeat, never on arm, so all arms see identical label subsets and the comparison is paired.

**Protocol parity.** This sweep uses the *same* probe protocol as Table 1: minibatches of 256, AdamW at $4{\times}10^{-4}$, weight decay 0.05, 50 epochs with 5 warmup and cosine decay, and the epoch selected on validation AUC. An earlier cheaper full-batch fit disagreed with the primary protocol. Under matched protocols, restricting this check to the null and CENTROID, their full-supervision results agree to within $0.0003$ ($0.8748$ against $0.8746$ for the null, $0.8856$ against $0.8855$ for CENTROID), so the curve below and the headline table describe one probe, not two. This two-arm bound is distinct from the four-arm bound reported in Section 5.4.

The observed mean gap is larger in the low-label probe refits (Table 11, Figure 7). At $5\%$ of labels ($n{=}300$) CENTROID leads the null by $+0.0496$ ($0.8335$ against $0.7839$), against $+0.0108$ at full supervision, more than four times the margin. At $1\%$ the gap is wider still ($0.7576$ against $0.6961$), though the standard deviations there ($0.0359$–$0.0368$ across arms) are large enough that we read the ordering but not its precise size.

The defective COVER arm (Appendix G) is the worst arm at every fraction, reaching only $0.6879$ at $1\%$ against $0.6961$ for the null. Without corrected-policy training, that deficit cannot be attributed to truncation rather than the other differences between these implementations.

Table 11: Label efficiency: frozen-probe test AUC against the fraction of the labelled training set used. All values measured, 5 repeats per fraction except at full data, which is deterministic.

| labels | $n$ | RANDOM | CENTROID | ENVELOPE | COVER |
|---|---|---|---|---|---|
| 1% | 60 | 0.6961 | 0.7576 | 0.7471 | 0.6879 |
| 5% | 300 | 0.7839 | 0.8335 | 0.8197 | 0.7484 |
| 10% | 600 | 0.8169 | 0.8490 | 0.8404 | 0.7743 |
| 25% | 1500 | 0.8497 | 0.8732 | 0.8654 | 0.8162 |
| 100% | 6000 | 0.8748 | 0.8856 | 0.8813 | 0.8586 |

## K Unavailable attribution evidence

Earlier drafts included occlusion curves and heat maps from fine-tuned models. Their archived images remain available, but the underlying per-volume numerical arrays and a complete normalization record are not retained in the evidence used for this submission. We therefore exclude those quantitative displays and do not use them to support spatial-consistency, laterality or masking-mechanism claims. This does not change the saved frozen-probe AUC comparisons. Recovering the producing arrays, checkpoint identities and plotting transforms would be necessary before restoring those analyses. No new attribution campaign was run to replace the missing evidence.

## L Clinical operating points and calibration

AUC is threshold-free, but a screening tool is deployed at a threshold. The threshold is selected on the RANDOM arm's *validation* split at a fixed target specificity and then applied unchanged to every arm on the test split: one shared threshold rather than one selected separately for each arm. This is a threshold-transfer analysis, not a clinical deployment evaluation. The achieved test specificity shows whether the threshold survives the shift. Cohort prevalence is $0.4887$.

Table 12: Operating points at matched epoch 100. One threshold, chosen on the RANDOM validation split to hit the target specificity, is applied unchanged to all arms on test. ECE is a 15-bin expected calibration error.

| policy | target spec. | sens. | spec. (test) | PPV | NPV | Brier | ECE |
|---|---|---|---|---|---|---|---|
| RANDOM | 0.85 | 0.7701 | 0.8259 | 0.8087 | 0.7899 | 0.1416 | 0.0355 |
| RANDOM | 0.90 | 0.7162 | 0.8794 | 0.8502 | 0.7643 | 0.1416 | 0.0355 |
| ENVELOPE | 0.85 | 0.7735 | 0.8351 | 0.8176 | 0.7942 | 0.1364 | 0.0361 |
| ENVELOPE | 0.90 | 0.7306 | 0.8807 | 0.8541 | 0.7738 | 0.1364 | 0.0361 |
| CENTROID | 0.85 | 0.7851 | 0.8318 | 0.8169 | 0.8020 | 0.1341 | 0.0320 |
| CENTROID | 0.90 | 0.7428 | 0.8696 | 0.8448 | 0.7797 | 0.1341 | 0.0320 |

Two observations (Table 12). First, the threshold transfers imperfectly: a validation target of $0.90$ yields test specificity $0.8696$ to $0.8807$ across the three arms, so the operating point moves by roughly one point between arms and the comparison below is at a shared threshold rather than at a shared false-positive rate. Second, sensitivity is higher at this shared threshold. At a target specificity of $0.90$, CENTROID detects $0.7428$ of cases against $0.7162$ for the unguided null, a paired difference of $+0.0266$ (CI $[+0.0136, +0.0396]$, excluding zero), with the specificity caveat above. At a target of $0.85$ the gain is $+0.0150$ (CI $[+0.0020, +0.0280]$). These four arm-by-target CIs use 10,000 class-stratified percentile subject resamples with fitted heads and thresholds fixed. They are one unadjusted exploratory family. CENTROID also has lower measured overall Brier score ($0.1341$ versus $0.1416$) and ECE ($0.0320$ versus $0.0355$). These are calibration point estimates on this split, without uncertainty or a reliability-curve comparison; they do not establish uniformly better calibration across subgroups.

A $+0.011$ AUC difference is hard to interpret clinically, and the ROC curves are nearly superimposed (Figure 8), whereas a change in cases detected at a fixed threshold is not. We state that comparison carefully: transferring one threshold moves achieved specificity from $0.8794$ to $0.8696$, so the arms are *not* compared at an identical false-positive rate and the sensitivity change partly reflects that shift. A deployment-grade comparison would select each arm's threshold on validation to meet the same specificity, lock it, and evaluate on an untouched cohort. One dataset, one split, one pretraining run per policy.

**Who receives the benefit at the deployed threshold.** Subgroup AUC is also threshold-free, so it cannot answer the question a deployment actually raises: a screen ships *one* threshold shared by everyone, and a model can lift every subgroup's AUC while delivering the gain unevenly at that single operating point. We therefore fix the threshold on validation at specificity $0.90$ ($0.5552$), transfer it unchanged, and measure the change in sensitivity within each stratum (Table 13). Overall sensitivity rises from $0.7162$ to $0.7428$.

The point estimates alone suggest the gain is concentrated in the largest group: $+0.0343$ for white patients against $+0.0037$ for black patients, an apparent ninefold difference. **The intervals do not**

**support that reading.** The black interval ($[-0.0375,\ +0.0449]$) reaches almost exactly the white point estimate, so the two are not separated. What distinguishes them is positive count ($1080$ versus $268$), not demonstrated benefit. Only the white, female and male gains exclude zero. All three remain after simultaneous adjustment over the five-contrast family, and they are the three largest strata. This is a *power* limitation of the cohort: the study cannot certify that the smaller strata benefit, and equally cannot show they do not. Certifying subgroup benefit at a deployed threshold would need a cohort with more positives in the smaller strata.

**The threshold itself does not transfer evenly across strata.** The same table answers a question the sensitivity column does not. A threshold selected on validation to achieve specificity $0.90$ does not achieve it uniformly on test: under RANDOM it realises $0.895$ specificity in the white stratum but $0.736$ in the black stratum, a shortfall of roughly sixteen points, and under CENTROID the same threshold realises $0.879$ and $0.761$. The shortfall is therefore a property of transferring one threshold to a cohort whose strata have different score distributions, and it is present under *both* arms. No masking policy here causes it and none of them repairs it. For a screening context this is a larger effect than any AUC difference in this paper.

**The sensitivity gain is not a uniform trade.** Reporting sensitivity alone would make the change from RANDOM to CENTROID look like pure benefit where it is in fact a trade. In the white stratum sensitivity rises by $+0.034$ while specificity falls by $-0.016$. In the black stratum specificity *rises* by $+0.025$ and in the asian stratum by $+0.008$, alongside sensitivity changes ($+0.004$ and $+0.008$) whose intervals contain zero. By gender, both strata gain sensitivity ($+0.026$ female, $+0.027$ male) and lose specificity ($-0.008$ and $-0.013$). Subgroup calibration moves the same way and not entirely in our favour: expected calibration error improves overall ($0.0355$ to $0.0320$) and in the white ($0.0382$ to $0.0321$) and asian ($0.0832$ to $0.0729$) strata, but *worsens* in the black stratum ($0.0525$ to $0.0569$). Race is the only attribute for which we report stratified calibration. All values in these two paragraphs are measured at the same shared threshold as Table 13, and the comparison remains at a shared threshold rather than at a shared realised false-positive rate.

Table 13: Change in sensitivity from RANDOM to CENTROID at one shared validation-selected threshold, by stratum. The five contrasts form one exploratory family. Reported CIs are 95% single-step max-$|t|$ simultaneous intervals from 10,000 paired positive-subject resamples shared across both fixed heads and all five overlapping race and gender strata, with heads and threshold fixed (family-wise $\alpha = 0.05$). White, female and male remain above zero after adjustment. Epoch 100.

| stratum | positives | $\Delta$ sensitivity | simultaneous 95% CI | excludes 0 |
|---|---|---|---|---|
| Asian | 118 | $+0.0085$ | $[-0.0485,\ +0.0654]$ | no |
| Black | 268 | $+0.0037$ | $[-0.0375,\ +0.0449]$ | no |
| White | 1080 | $+0.0343$ | $[+0.0149,\ +0.0536]$ | yes |
| Female | 811 | $+0.0259$ | $[+0.0020,\ +0.0498]$ | yes |
| Male | 655 | $+0.0275$ | $[+0.0043,\ +0.0507]$ | yes |

## M Broader impact and ethics

This study uses a public, de-identified retinal dataset (FairVision) under its release terms and involves no new recruitment or data collection. No model reported here is clinically validated or intended for deployment: the frozen-probe AUCs are research measurements on one dataset and one task, and are not evidence of clinical utility.

**Dataset provenance and licence.** We name the governing terms explicitly rather than leave "release terms" to stand on its own. The cohort is the glaucoma arm of Harvard-FairVision [Luo et al., 2024a]: 10,000 subjects distributed with a fixed 6,000/1,000/3000 training, validation and test partition. The primary downstream comparisons use the 3000-subject test partition; secondary diagnostic subsets are identified separately. The distribution states a data licence of CC BY-NC-ND 4.0, which permits non-commercial research and states that the data are not to be used for clinical decisions or patient care. The accompanying code repository carries a separate MIT software licence that covers code only and does not relax the data terms. Our use is non-commercial research measurement with no clinical deployment, which is what those terms allow. The data were taken from the public release. No application, data use agreement or credentialed-access step is recorded for this dataset in our project record. The release is distributed de-identified: each subject archive holds an OCT volume, a binary label, a visual-field mean deviation and coded demographic attributes, with no name, date or free text. We added no identifying field, attempted no re-identification, and linked to no external source. The reproduction artifacts are encoder weights, a fitted probe head and per-case prediction scores. The submission contains no raw OCT volumes.

**What our ethics record does not contain.** This project holds no IRB or ethics-committee approval number, no consent documentation, and no written non-human-subjects or exemption determination for this secondary analysis. None was recorded at the time, and we will not assert one after the fact. The statement above is therefore narrower than a governance statement: it is what the release says and what we did with it. The approval and consent record for the underlying data collection belongs to the FairVision authors' primary publication and the institutional record behind it [Luo et al., 2024a], which we did not obtain. Closing this gap would require citing that ethics statement directly and obtaining a written determination from our own institution. A venue requiring a formal determination should treat this item as outstanding rather than satisfied.

The subgroup analysis in Section 5.5 is a caution, not a contribution. For five of seven attributes the same eligible group is worst-served across the retained probes. The highlighted race, gender and disease-severity point estimates rise at the matched epoch, but this does not hold for every recorded attribute: the divorced group's point estimate decreases. Paired differential-benefit tests cover specified contrasts, not all seven attributes, and do not establish a general reduction in disparities. We would regard deployment of any encoder reported here, without a dedicated and prospective fairness evaluation on the target population, as inappropriate. We also stress that the audit is incomplete: beyond AUC, the transferred-threshold sensitivity, specificity and calibration of Appendix L (whose expected calibration error is stratified by race, the only attribute for which we report it) and the race-by-gender breakdown of Appendix E.1, it contains no predictive-value analysis, no reliability curves, and no calibration stratified by gender, ethnicity or severity, and the intersectional cells carry no paired per-cell intervals, so no per-cell change there is tested against zero.

Releasing pretrained weights carries the usual dual-use consideration that they may be applied to populations unlike the training cohort. We document the cohort composition (Appendix E) so that such a mismatch is detectable by anyone reusing the weights. The demographic attributes are self-reported categories inherited from the dataset's coding. They are used only for the subgroup audit and are never model inputs.

# N Published ablations on informed versus random masking

Random masking is a strong baseline in the published record and informed masking wins only under some objectives and protocols, the deficit largest under frozen evaluation. Our result should be read against those comparisons (Table 14), collected while positioning this work.

Read together, these say that the *direction* of our finding is not new. What we add is an audited run-level and delivered-task measurement, not a general claim about the sign of the masking effect.

# O Full paired-contrast table

Benjamini–Hochberg is applied within a declared primary family consisting of exactly these nine contrasts (the fp16 comparisons among RANDOM, CENTROID and ENVELOPE at epochs 50, 75 and 100) and the remaining 25 exploratory contrasts in the stored statistics are corrected as a separate family. The two families are never pooled, and the $q$ values printed above are those of the nine-contrast family only. Declaring the family fixes what is corrected together. It does not make these contrasts confirmatory, because the test split was inspected repeatedly before the family was fixed, and the paper reports them as descriptive throughout.

# P Full fp32 re-probe

The RANDOM, CENTROID and ENVELOPE long-horizon probes were originally fitted under the harness default, which enables autocast. To assess whether the headline effect is a precision artifact,

Table 14: Published ablations comparing a random or simple baseline against an informed or structured masking scheme. Numbers are as reported by the cited papers; metrics differ across rows and are not comparable between rows.

| study | baseline | informed variant | interpretation |
|---|---|---|---|
| MAE [He et al., 2022] | random-75: 84.9 FT / 73.5 lin | block-75: 82.8 / 63.9; grid-75: 84.0 / 66.0 | random decisively better |
| SimMIM [Xie et al., 2022] | 83.0 top-1 | best block 82.7; square 82.6 | random modestly better |
| SemMAE [Li et al., 2022] | random 66.8 lin | within-part 66.5; whole-part 52.9; adaptive 68.7 | naive semantics hurt; scheduled semantics help |
| SemMAE part quality [Li et al., 2022] | no parts 63.7 | raw parts 63.6; refined 65.0 | an imperfect semantic model adds nothing |
| AutoMAE [Chen et al., 2023a] | MAE 83.26 FT | AutoMAE 83.32 | effectively tied |
| HPM [Wang et al., 2023] | random 82.49 | moderate 82.95; hardest-only 81.40 | benefit only with retained randomness |
| AttMask [Kakogeorgiou et al., 2022] | random 43.4 $k$-NN | high 43.5; hint 43.6 | nearly indistinguishable |
| AttG-MAE [Sick et al., 2025] | MAE 78.5 @10% labels | guided 78.1 | guidance loses in this protocol |
| SSiT [Huang et al., 2024] | no saliency 79.48 $\kappa$ | 25% mask 77.53; 50% 79.97 | dataset-dependent, non-monotonic |

Table 15: Primary contrasts: matched epoch *and* matched precision. $\Delta$ is a paired difference on identical test cases; the interval is a 10,000-resample paired bootstrap and $p$ is a DeLong test for correlated ROC curves. $q$ is Benjamini–Hochberg over these nine contrasts. All $p$ and $q$ values here are descriptive (Section 6). Every comparison against the null excludes zero.

| contrast | epoch | $\Delta$ AUC | 95% CI | $p$ | $q$ |
|---|---|---|---|---|---|
| ENVELOPE − RANDOM | 50 | +0.0120 | [+0.0068, +0.0173] | <0.0001 | <0.0001 |
| ENVELOPE − RANDOM | 75 | +0.0080 | [+0.0028, +0.0132] | 0.0025 | 0.0045 |
| ENVELOPE − RANDOM | 100 | +0.0062 | [+0.0010, +0.0113] | 0.0199 | 0.0299 |
| CENTROID − RANDOM | 50 | +0.0099 | [+0.0051, +0.0149] | <0.0001 | 0.0001 |
| CENTROID − RANDOM | 75 | +0.0113 | [+0.0063, +0.0164] | <0.0001 | <0.0001 |
| CENTROID − RANDOM | 100 | +0.0109 | [+0.0057, +0.0160] | <0.0001 | <0.0001 |
| CENTROID − ENVELOPE | 50 | -0.0020 | [-0.0069, +0.0029] | 0.4114 | 0.4114 |
| CENTROID − ENVELOPE | 75 | +0.0033 | [-0.0013, +0.0079] | 0.1491 | 0.1677 |
| CENTROID − ENVELOPE | 100 | +0.0047 | [+0.0004, +0.0091] | 0.0294 | 0.0378 |

we re-ran the complete probe pipeline for those arms with autocast disabled, so that feature extraction, head optimisation and test evaluation are all fp32, under the exact protocol already used by the ANATOMY and COVER arms (Table 16). The encoder checkpoint is SHA-256 hashed before and after each run and verified unchanged, so no result here can be contaminated by an accidental encoder update.

# Q Reproducibility and numeric provenance

**Software and hardware environment.** The local analysis and engineering-diagnostic environment was recorded as Python 3.11.9 on 64-bit Windows (build 10.0.26200); PyTorch 2.7.1 compiled against CUDA 12.8 with cuDNN 9.7.1; one NVIDIA GeForce RTX 3090 with 24 GiB of memory, driver 610.62. The statistics and figures use NumPy 2.4.4, SciPy 1.17.1 and scikit-learn 1.9.0, and the manuscript is compiled by Tectonic 0.17.0. A lock file pinning 87 distributions accompanies the artifact. At the analysis snapshot those pins matched the installed versions. Additional paper-review tooling is not a reconstruction of every historical training host. The chain from stored per-case predictions through macros, tables and figures to the compiled archive runs as one ordered script, so the retained predictions, the retained head and the numbers printed here are joined by code rather than by transcription. We did not repeat every training and evaluation run under a second CUDA or

Table 16: Full fp32 re-probe against the original fp16 result, same frozen encoder in each row. The eight DeLong $p$-values are unadjusted diagnostics, not a family-wise test. The largest difference is $2 \times 10^{-4}$.

| policy | epoch | fp16 AUC | fp32 AUC | $\Delta$ | $p$ |
|---|---|---|---|---|---|
| ENVELOPE | 50 | 0.876064 | 0.876063 | -0.000001 | 0.964 |
| ENVELOPE | 75 | 0.880307 | 0.880305 | -0.000002 | 0.806 |
| ENVELOPE | 100 | 0.880743 | 0.880761 | +0.000018 | 0.167 |
| CENTROID | 50 | 0.874030 | 0.874015 | -0.000015 | 0.412 |
| CENTROID | 100 | 0.885485 | 0.885293 | -0.000192 | 0.767 |
| RANDOM | 50 | 0.864097 | 0.864121 | +0.000024 | 0.128 |
| RANDOM | 75 | 0.872302 | 0.872302 | -0.000001 | 0.962 |
| RANDOM | 100 | 0.874581 | 0.874485 | -0.000096 | 0.000 |

PyTorch version. The fp32 re-probes and fixed-head cached-feature check have the narrower scopes described in Section 5.6.

**Why the intervals are non-parametric.** The estimator choice was checked against the data rather than assumed. The per-case paired differences are strongly non-normal with heavy tails (Shapiro–Wilk $W = 0.968$–$0.981$, $p < 10^{-10}$), and the raw within-arm scores more so ($p \approx 10^{-37}$). Levene's test also rejects equal variance across arms ($F = 21.4$, $p < 0.001$), though the variance ratio is only $1.16$ and is detectable mainly because $n = 9{,}000$. None of this invalidates what we report: a percentile bootstrap makes no distributional assumption about the deltas, which is the reason for choosing it, and an AUC is rank-based, so DeLong's variance is built from placement values rather than from Gaussian scores. The diagnostics are recorded here because "the intervals are non-parametric" is a claim a reader should be able to check, not take on trust. Where $n$ is small (the five-arm and four-arm correlations), a normality test has essentially no power, so we treat those as descriptive throughout and draw no inference from them.

**Guide provenance differs across the segmenter arms.** The three segmenter-driven policies do not all read the same anatomy guide. ENVELOPE uses the MIRAGE-derived repaired envelope without the residual adapter. ANATOMY-V1 and COVER instead read pre-computed *soft* guides produced by a MIRAGE whose segmentation path carries a frozen residual adapter, trained label-free to align MIRAGE's patch-similarity structure with the pretraining teacher's. That adapter was an abandoned line of work: it reduced its own training objective but never improved segmentation, and the experiment that would have settled its downstream effect was not run, so we make no claim about it here. We record it because it is a further axis on which the anatomy arms and COVER differ from ENVELOPE, beyond the geometric axes in Section 5.2, and because the guide directory names embed the adapter configuration and would otherwise be unexplained to anyone reproducing this work. CENTROID and RANDOM consult no guide of either kind.

**What is checked mechanically.** Generated result quantities are compared with named source statistics. Protocol constants, literature values and retained illustrations have separate source-review obligations. Resolving macros and compiling a package do not establish scientific correctness or prevent every inconsistency. Release checks cover source bindings, references, declared assets and document structure, with unresolved items distinguished from verified values.

**Checkpoint identity.** The shared epoch-25 ancestor is identified by SHA-256 (`e5ad5b0c2aadfa15449409786afbfa39d8b5405b699be8f02f2e540195e97e7b`), which was checked against the retained copies, and the replication chain of Appendix B re-hashes it before each leg. The frozen-probe implementation disables encoder gradients; file hashes bind the saved checkpoint but are not, by themselves, a check of every in-memory parameter update. The six proposed replication configurations differ in policy, seed and logging fields, not only seed.

**Unretained evidence.** An earlier per-cell concentration claim and probe-seed noise bound had no retained producing artifacts and are not reported as results here. The corresponding spatial-consistency mechanism is not established (Sections 5.2 and 6). Provenance and the appropriateness of the analysis are necessary for a numerical claim, whether the value is written directly or generated.

## R Fine-tuning narrows the observed gap

Under full fine-tuning rather than a frozen probe, CENTROID reaches $0.8947$ (mean-pool head) against RANDOM's $0.8868$, a gap of $+0.0079$ compared with $+0.0109$ frozen. Taking each arm's best head instead, the gap is $+0.0069$. The observed gap is smaller after supervised adaptation. These test-set-selected head point estimates do not distinguish representation quality from optimisation. These runs used a different head and schedule from the frozen probes and are never pooled with them (Table 17).

Table 17: Full fine-tuning, same test split: every arm/head combination, ordered by test AUC. Head comparisons are descriptive point estimates, reported separately from frozen probes and never pooled with them.

| policy | head | test AUC |
|---|---|---|
| CENTROID | mean-pool | 0.894668 |
| CENTROID | cross-attention | 0.893717 |
| CENTROID | attentive | 0.890100 |
| RANDOM | attentive | 0.887763 |
| RANDOM | cross-attention | 0.887178 |
| RANDOM | mean-pool | 0.886756 |

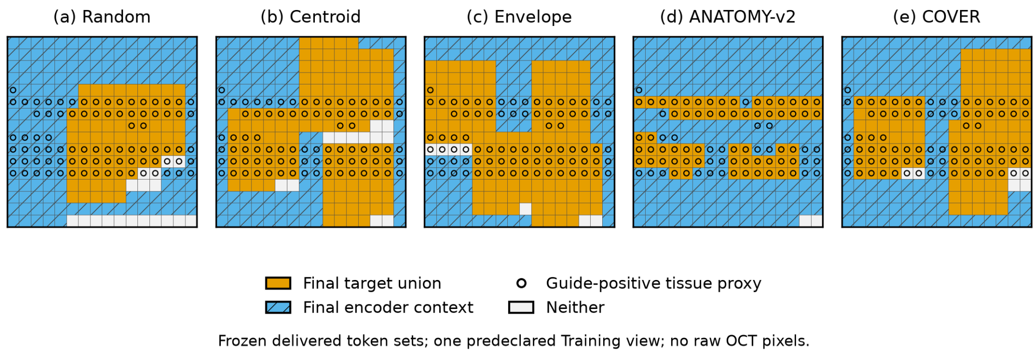


Figure 4: Delivered token sets for five policy families on the first predeclared Training view in the frozen two-view engineering-audit fixture. Orange cells are the union of final predictor targets, hatched blue cells are final encoder context, and hollow circles mark occupancy-guide-positive cells; other cells are neither target nor context. The guide is a tissue proxy, not clinical ground truth. All panels use the same input view and the already stored masks, without redrawing seeds or displaying raw OCT pixels. ANATOMY-v2 represents the shaped family, not ANATOMY-v1; COVER shows the historical implementation, not the proposed correction. Placements are not exactly paired across policies, and unions do not show repeated loss slots. This selected engineering example is not a population summary or evidence of downstream performance.

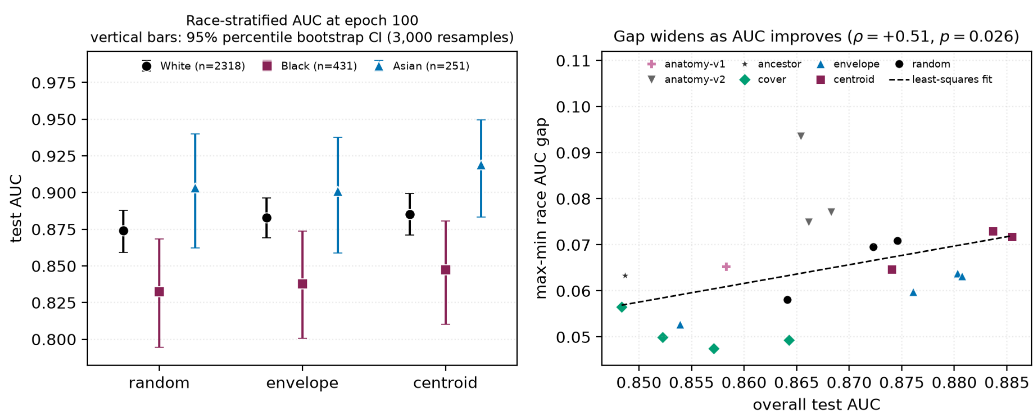


Figure 5: **Left:** race-stratified AUC at epoch 100. Points are group AUCs and vertical bars are 3,000-resample class-stratified percentile bootstrap intervals within each group, with fitted heads fixed and per-group $n$ in the legend. The black subgroup is lowest under every policy, yet *every* group point estimate rises from RANDOM to CENTROID. **Right:** max–min race gap against aggregate AUC, one point per probe over the 19 probes that carry a race summary; the legend keys colour and marker to policy. The dashed line is a least-squares fit while the quoted correlation is Spearman. The slope is positive but fails multiplicity correction at checkpoint level and does not survive branch-level aggregation (Table 6), so we draw no trend conclusion from it.

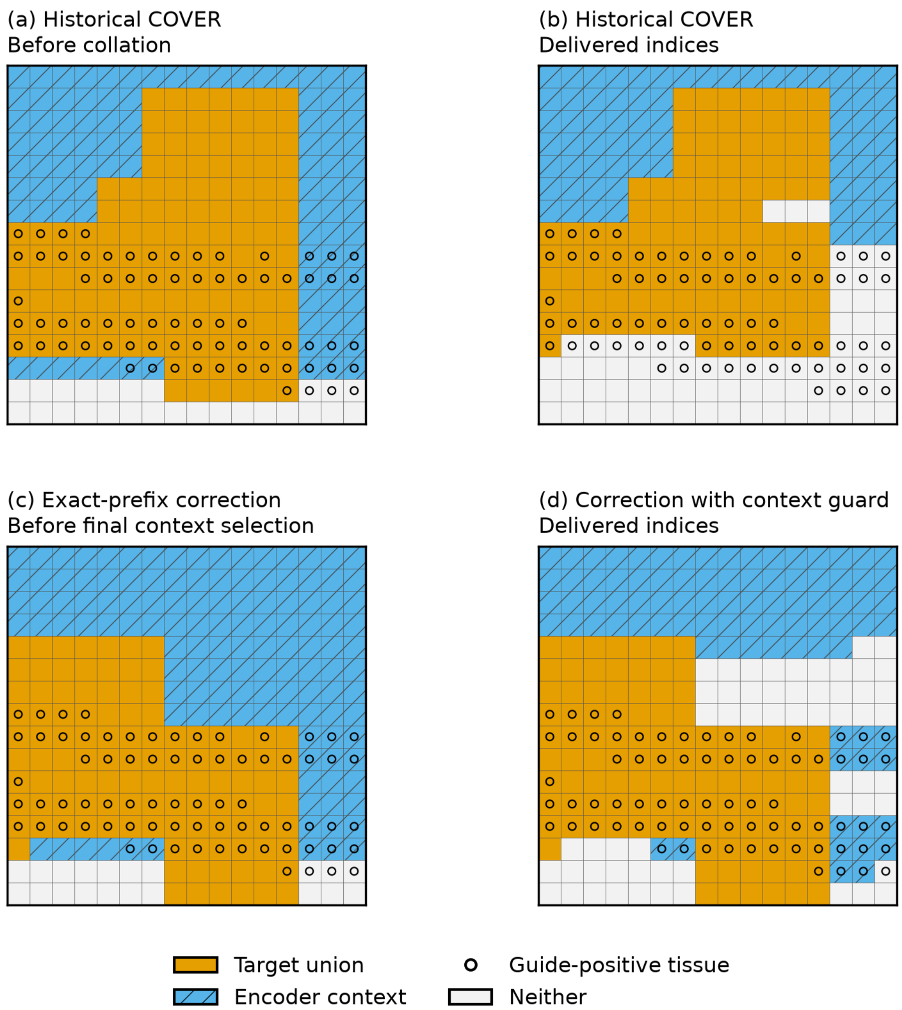


Figure 6: Intended and delivered token sets for one previously identified Training-view failure, not a randomly selected example. Orange cells are target unions, hatched blue cells encoder context, and circles guide-positive tissue. No raw OCT pixels are displayed. Historical collation removes guide-positive targets and all guide-positive tissue from the encoder context in this case. Exact-prefix scoring and the separate context guard change the delivered sets; the guard may select outside the candidate context rectangle. Input and drawn sizes are shared, but historical and corrected placements are not exactly paired. Aggregate evidence is in Table 10; this illustration does not imply a downstream AUC improvement.

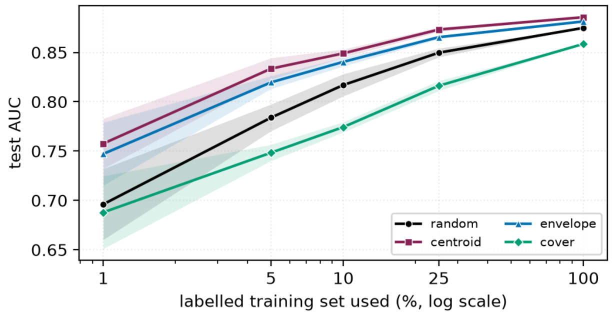


Figure 7: Frozen-probe results across label fractions. Shading is one standard deviation over 5 label subsets, with every arm seeing identical subsets. The full-supervision four-arm AUC spread is approximately $0.027$; at $5\%$ the spread is $0.085$.

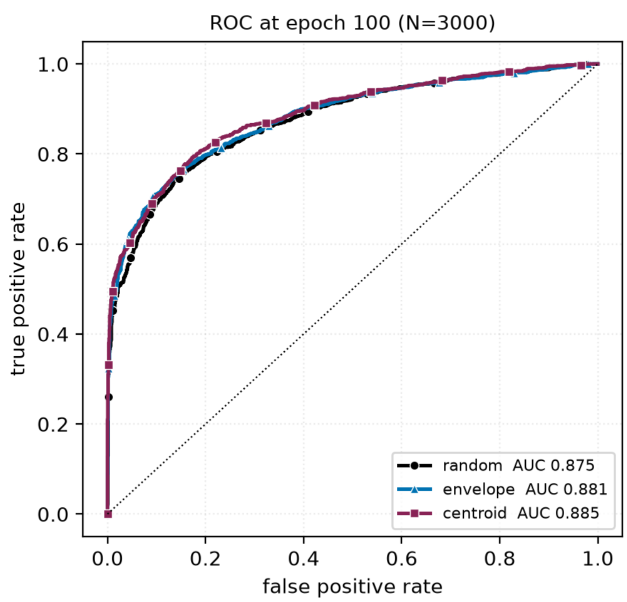


Figure 8: ROC curves at epoch 100 on the 3000 test volumes. The three arms plotted are nearly superimposed: the separation reported throughout this paper is a small, consistent shift in the high-specificity region rather than a difference visible in the curve as a whole. We include this so the effect size is not overstated by tables alone.

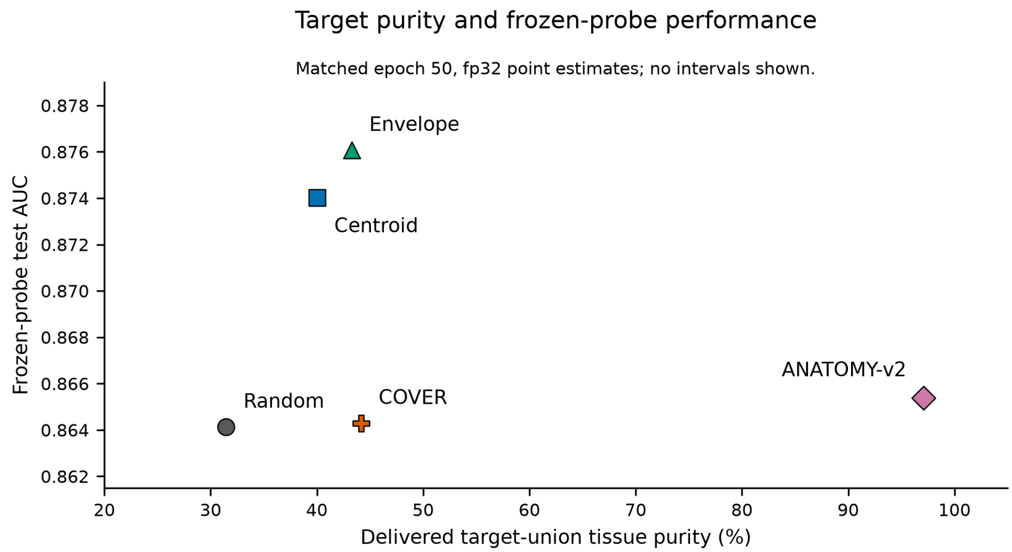


Figure 9: Delivered target-union tissue purity and frozen-probe test AUC for five policy families. Purity uses the current Table 2 geometry audit; all AUC points use epoch-50 fp32 probes. Centroid is the artifact arm named oracle, and ANATOMY-v2 is the shaped-policy representative; ANATOMY-v1 is not shown. Points are descriptive estimates from the evaluated continuations, with no fitted line or uncertainty intervals. The AUC axis does not start at zero. These confounded implementations do not isolate the effect of tissue purity or establish a policy ranking over retrainings.

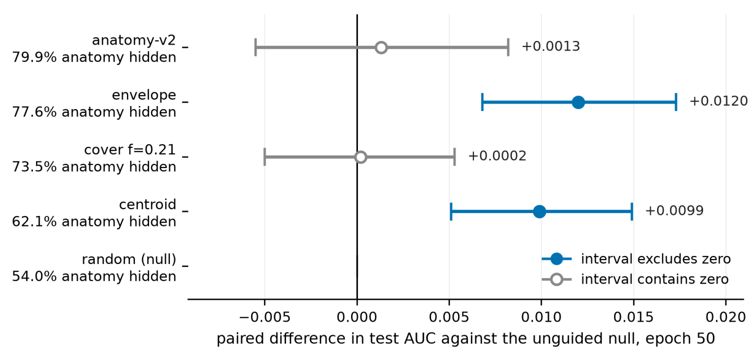


Figure 10: The matched-epoch-50 result arranged as a ladder of increasing anatomical specificity, measured as the share of tissue each policy's targets hide (Table 2). Points are paired differences in test AUC against the unguided null on identical cases, with 10,000-resample percentile bootstrap intervals. The axis is centred on zero, a meaningful baseline for a difference, with the displayed range labeled explicitly. Filled markers denote intervals excluding zero. The two arms that separate from the null are not the two that hide the most anatomy, which is the observation Section 5.2 develops. The intervals concern paired differences, not individual-arm AUC uncertainty. Companion to Figure 9.

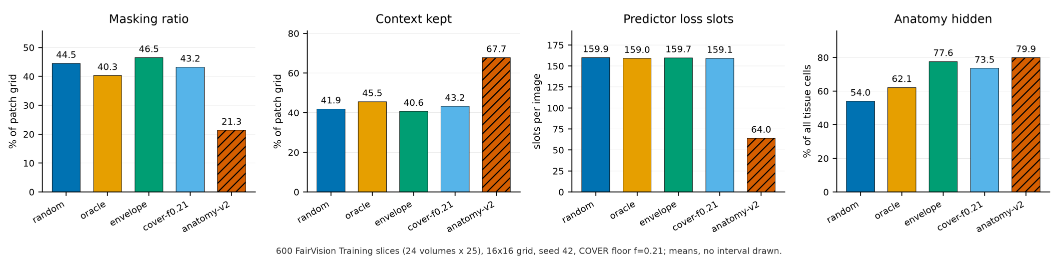


Figure 11: Measured mask geometry per policy, 600 training slices. Bars start at zero and no interval is drawn; each is a mean over those slices. The four rectangle policies have similar nominal target settings, but differ in realized masking ratio and context as well as anatomy hidden. The anatomy policy differs on all four axes (half the masking ratio, $1.6\times$ the context, $0.4\times$ the predictor loss slots), so its deficit cannot be attributed to target shape. Bars labelled `oracle` and `cover-f0.21` are CENTROID and COVER.